



US 20250285750A1

(19) **United States**

(12) **Patent Application Publication**
Ramesh et al.

(10) **Pub. No.: US 2025/0285750 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **SYSTEM AND METHODS FOR ENABLING
REMOTE PLANNED MAINTENANCE OF
MEDICAL IMAGING DEVICES**

Publication Classification

(51) **Int. Cl.**
G16H 40/40 (2018.01)

(52) **U.S. Cl.**
CPC G16H 40/40 (2018.01)

(57) **ABSTRACT**

One or more systems, devices, computer program products and/or computer-implemented methods of use provided herein relate to a process to facilitate remote planned maintenance of medical imaging devices. A system can comprise a memory that stores computer executable components, and a processor that executes the computer executable components stored in the memory, wherein the computer executable components can comprise an input component that receives sensor data from at least one medical device, and a degradation component that utilizes a feature transformation algorithm to determine a degradation score for a planned maintenance task of the at least one medical device based on the sensor data. Additionally, the computer executable components can comprise a planned maintenance component that recommends maintenance for the at least one medical device based on the degradation score.

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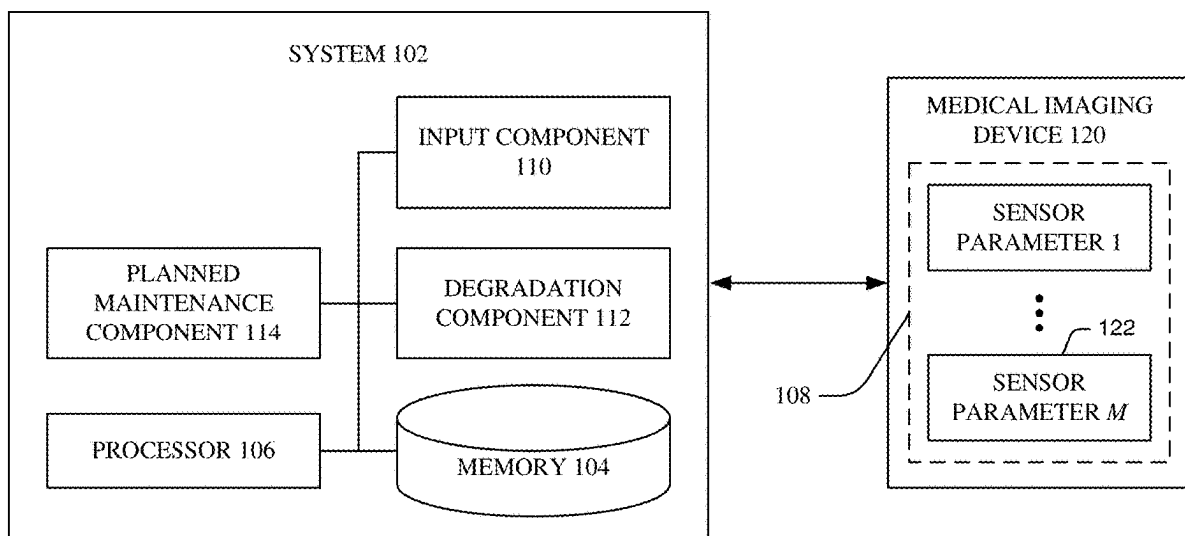
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(21) Appl. No.: **19/217,191**

(22) Filed: **May 23, 2025**

Related U.S. Application Data

(63) Continuation-in-part of application No. 18/057,347,
filed on Nov. 21, 2022.



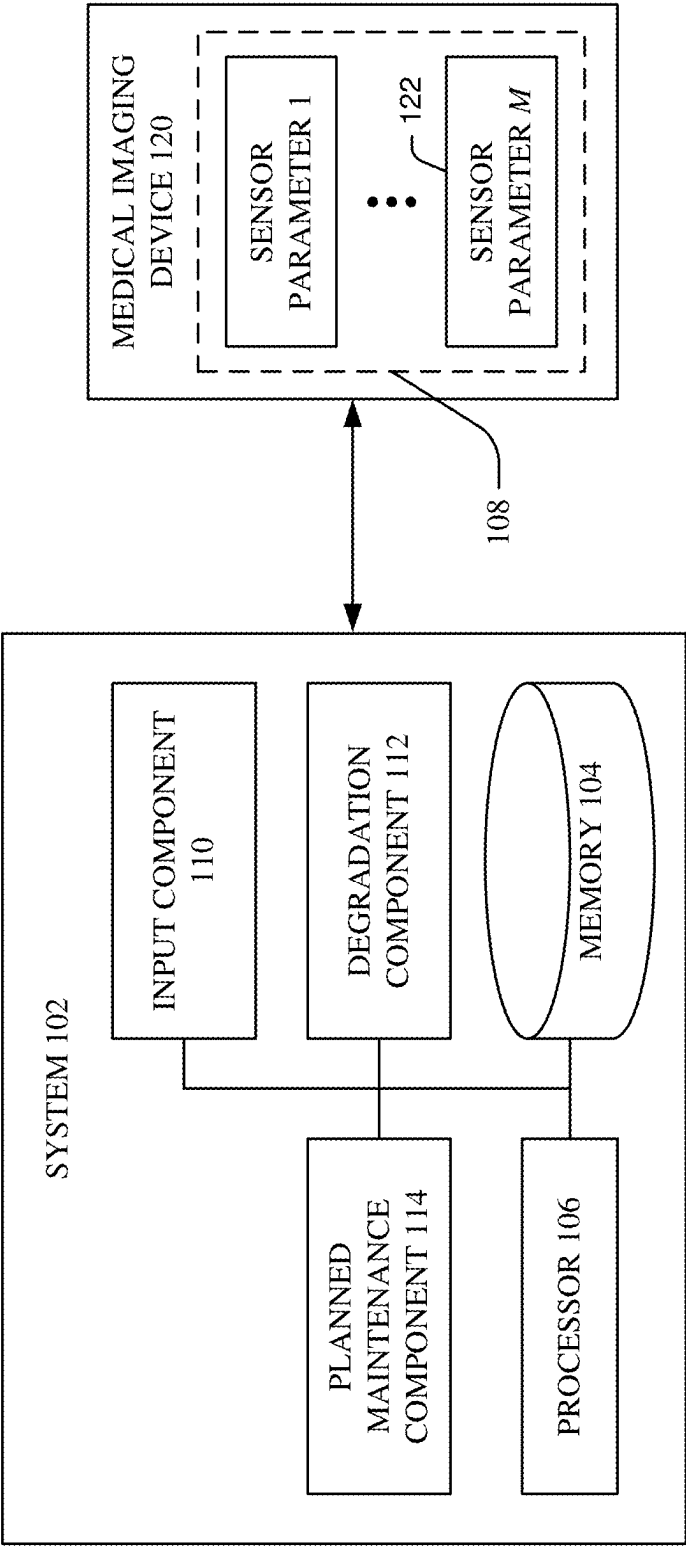


FIG. 1

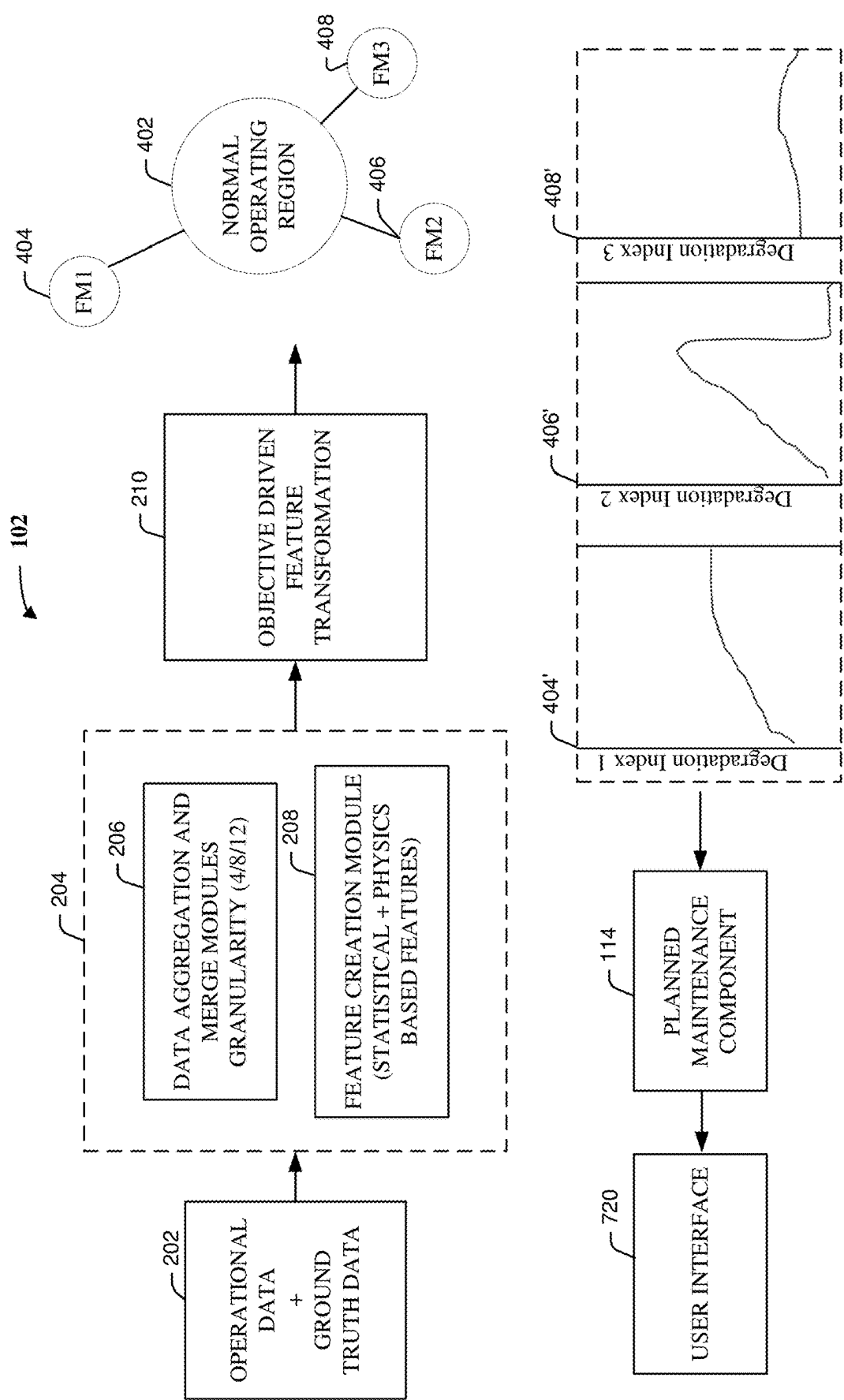
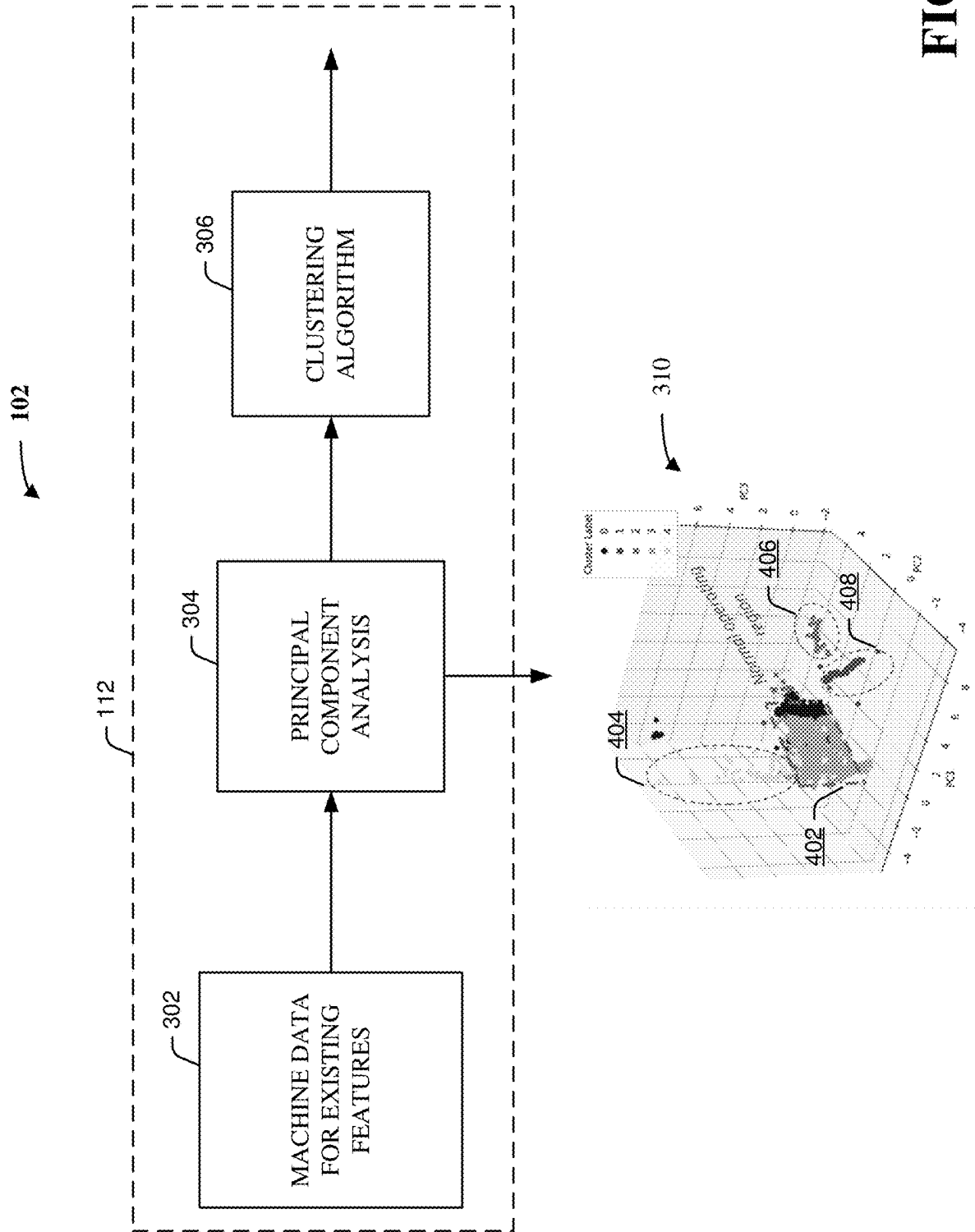


FIG. 2



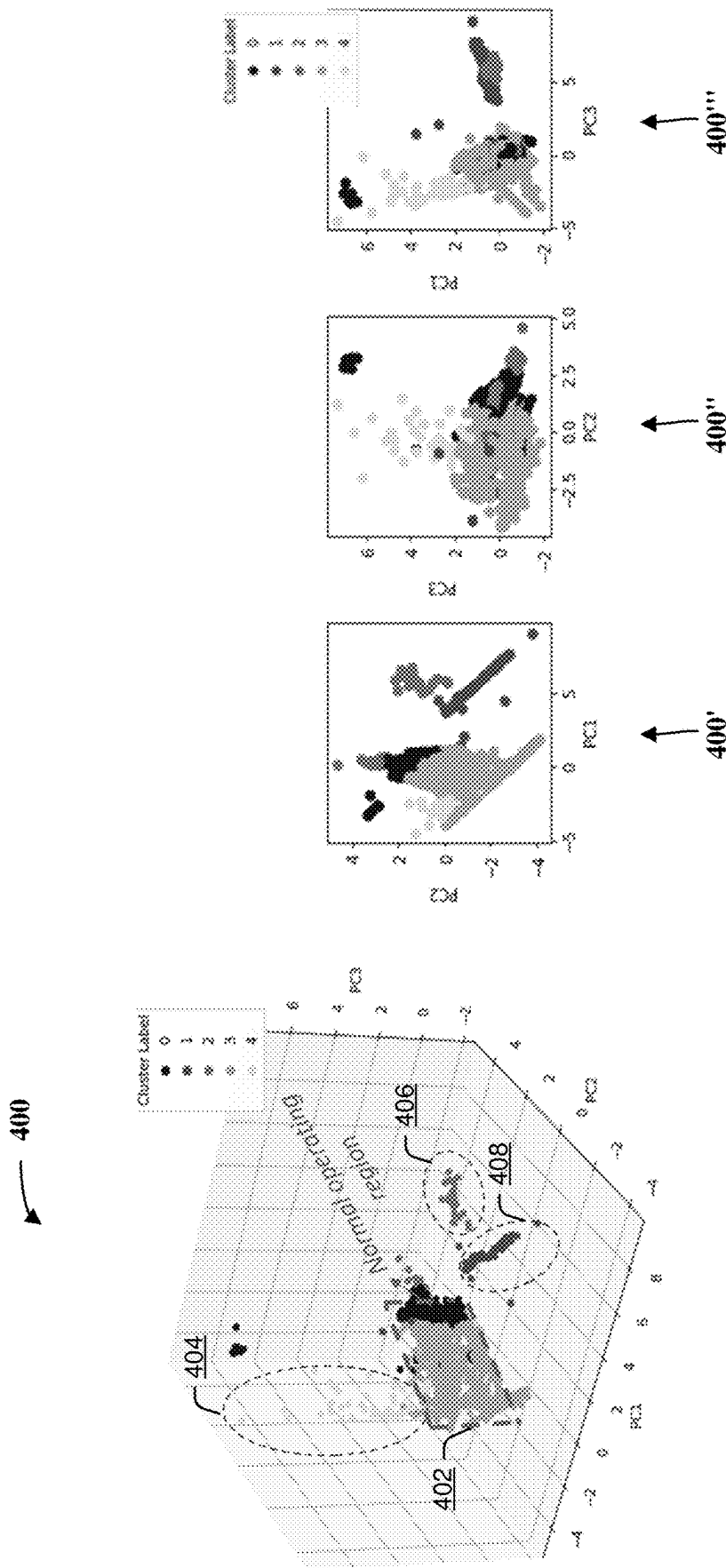


FIG. 4B

FIG. 4A

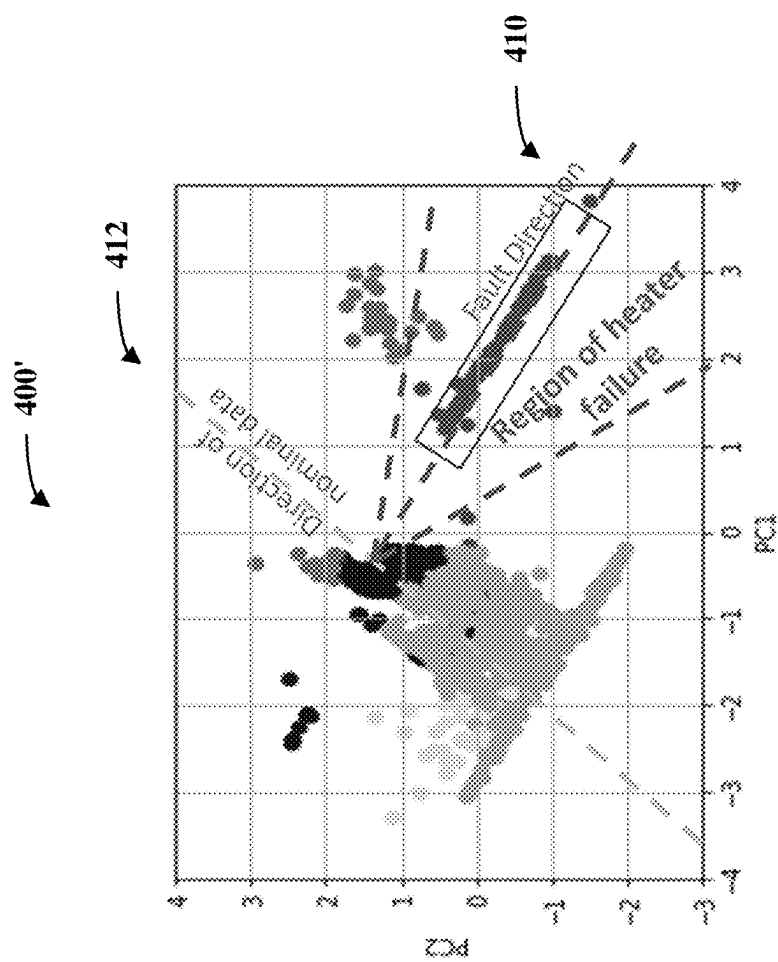


FIG. 4C

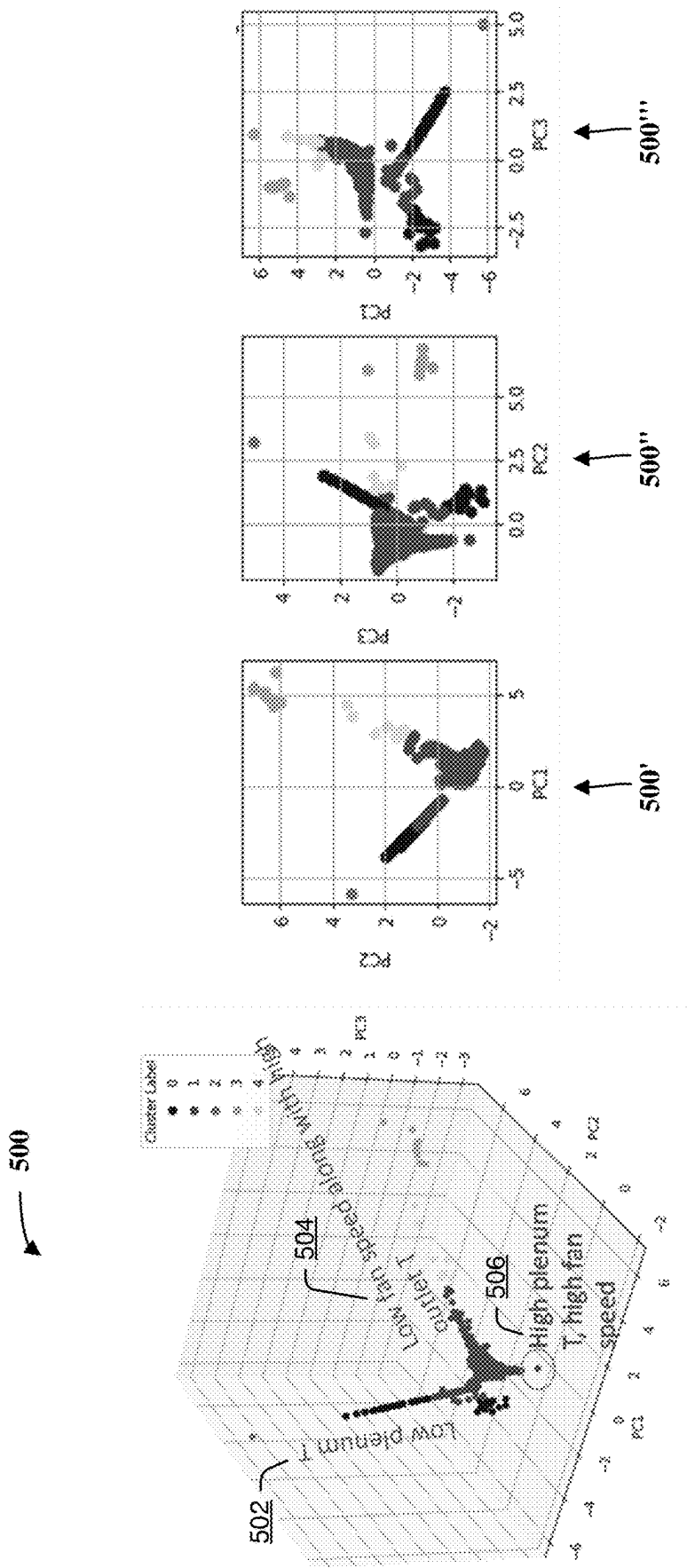


FIG. 5A

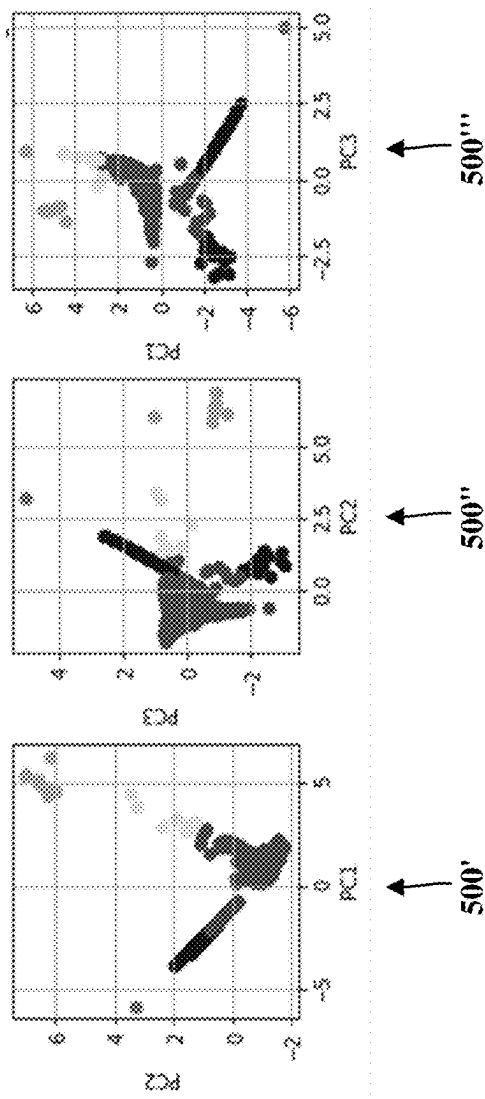


FIG. 5B

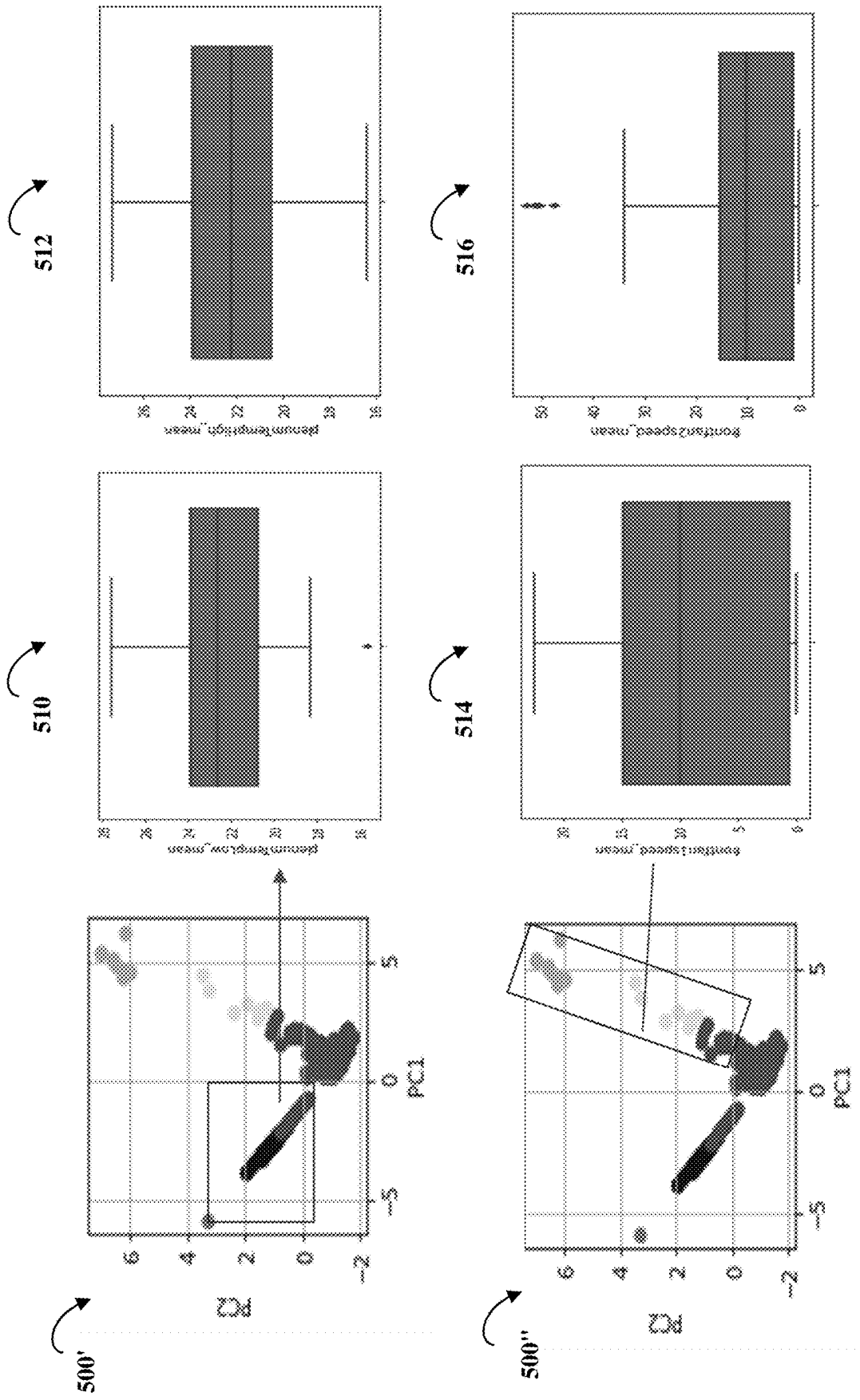
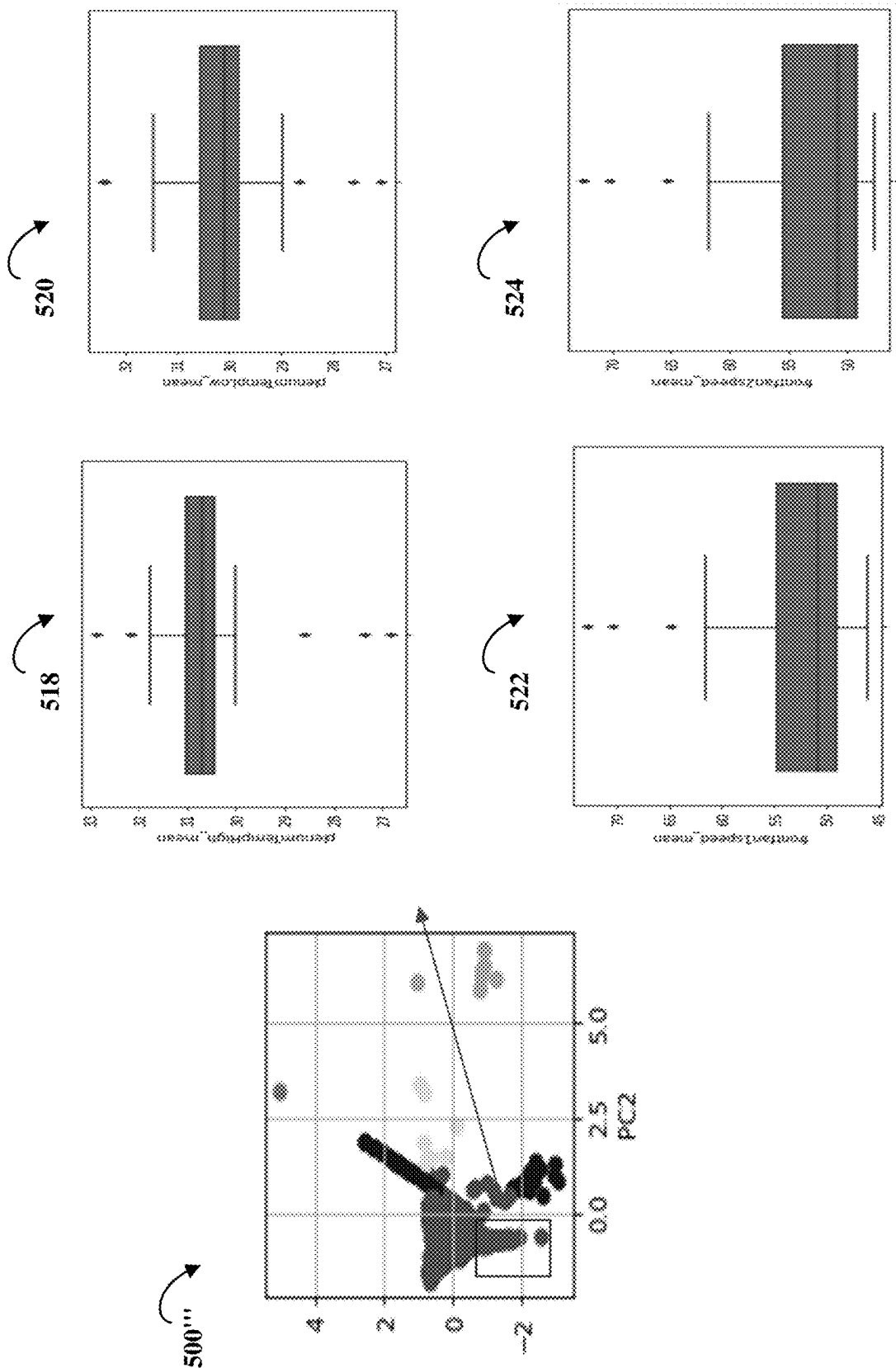


FIG. 5C



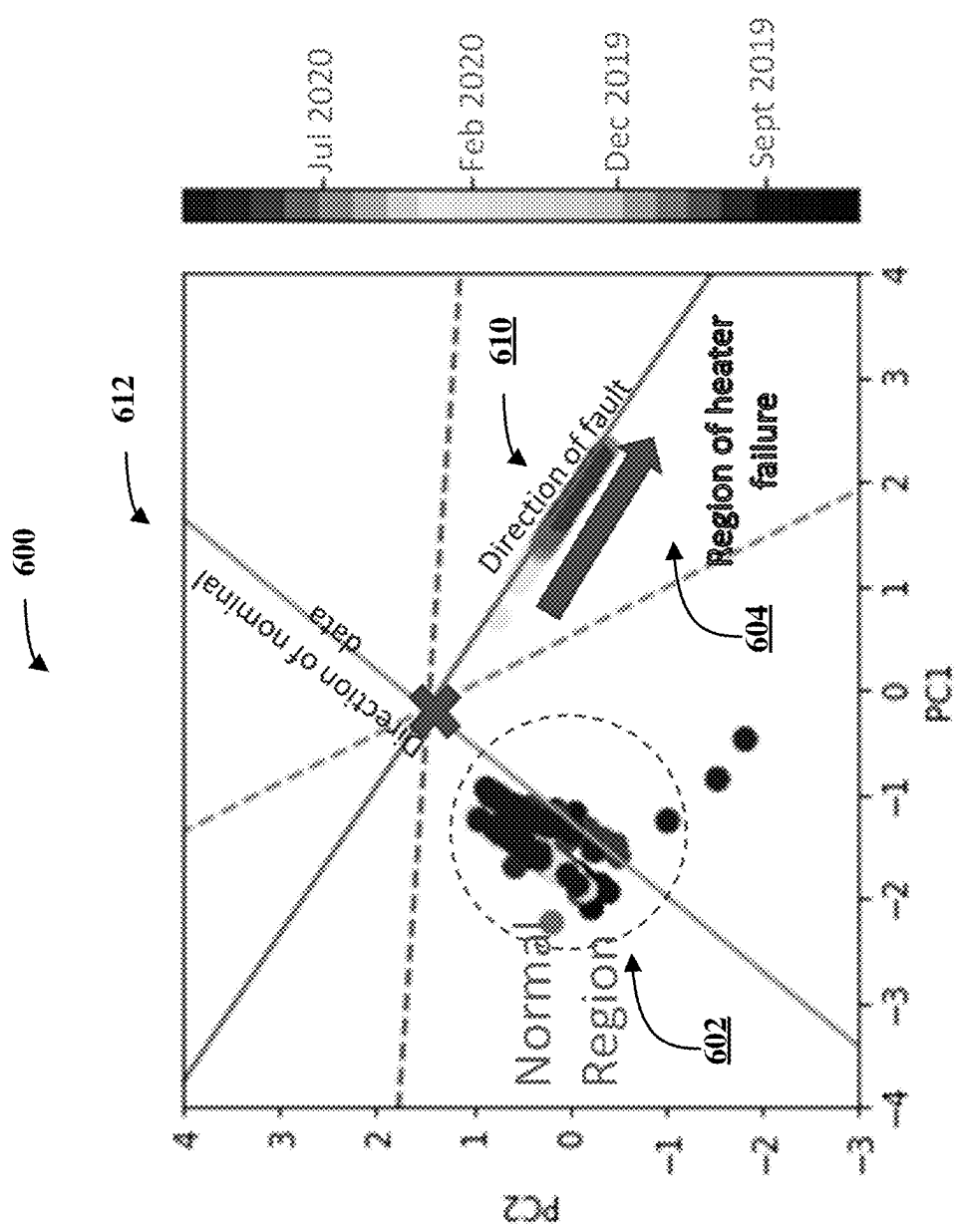


FIG. 6A

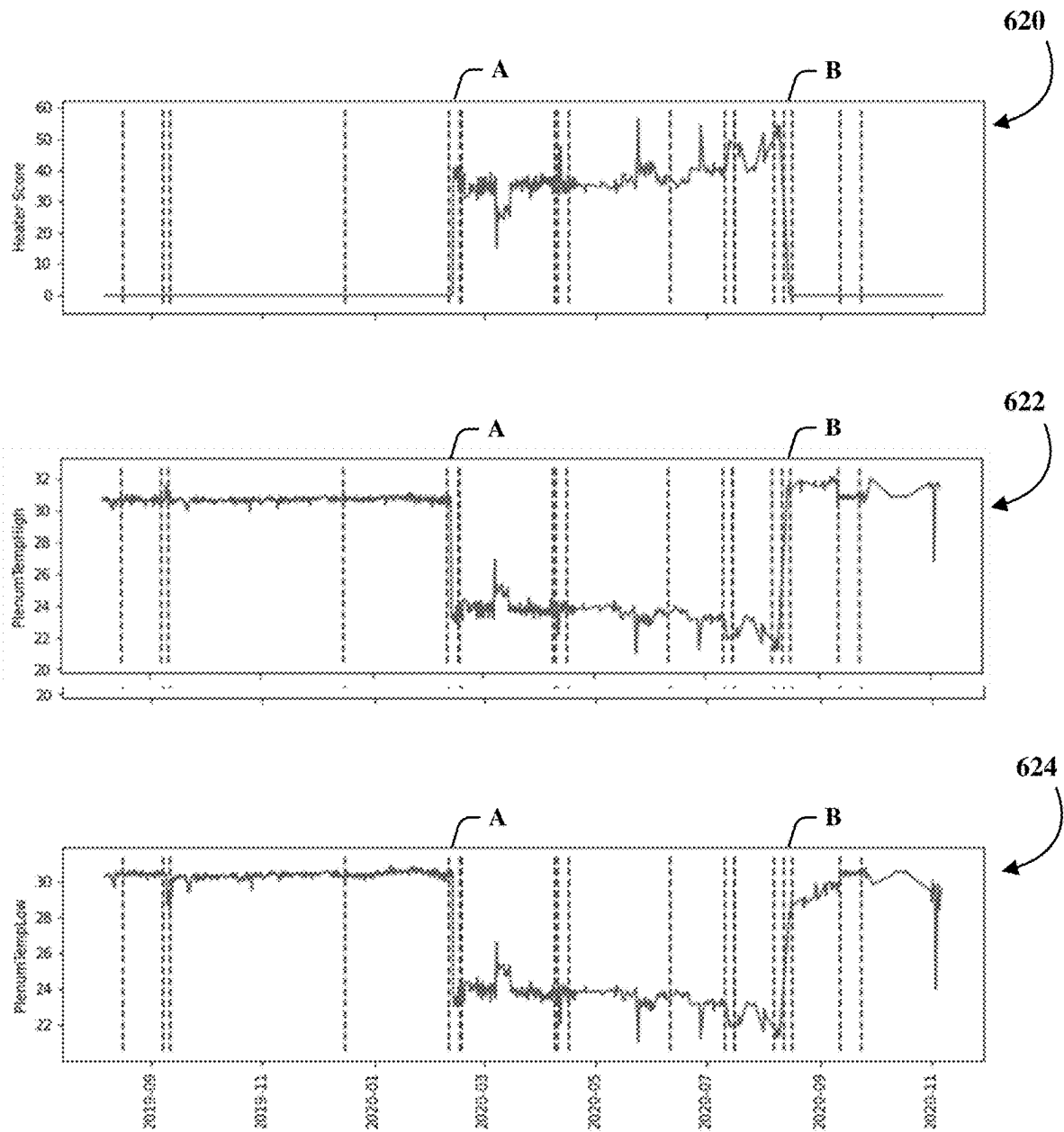
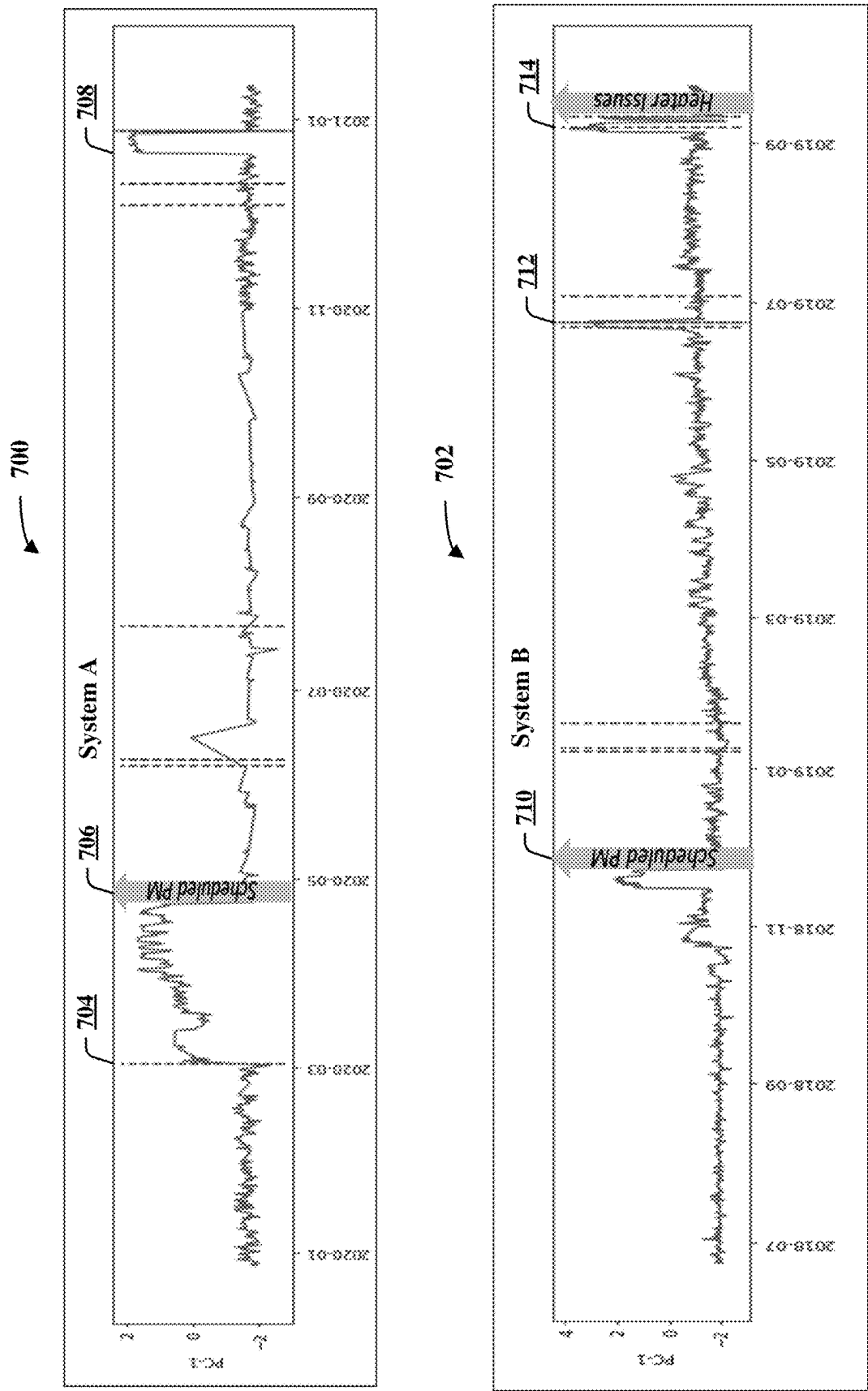


FIG. 6B



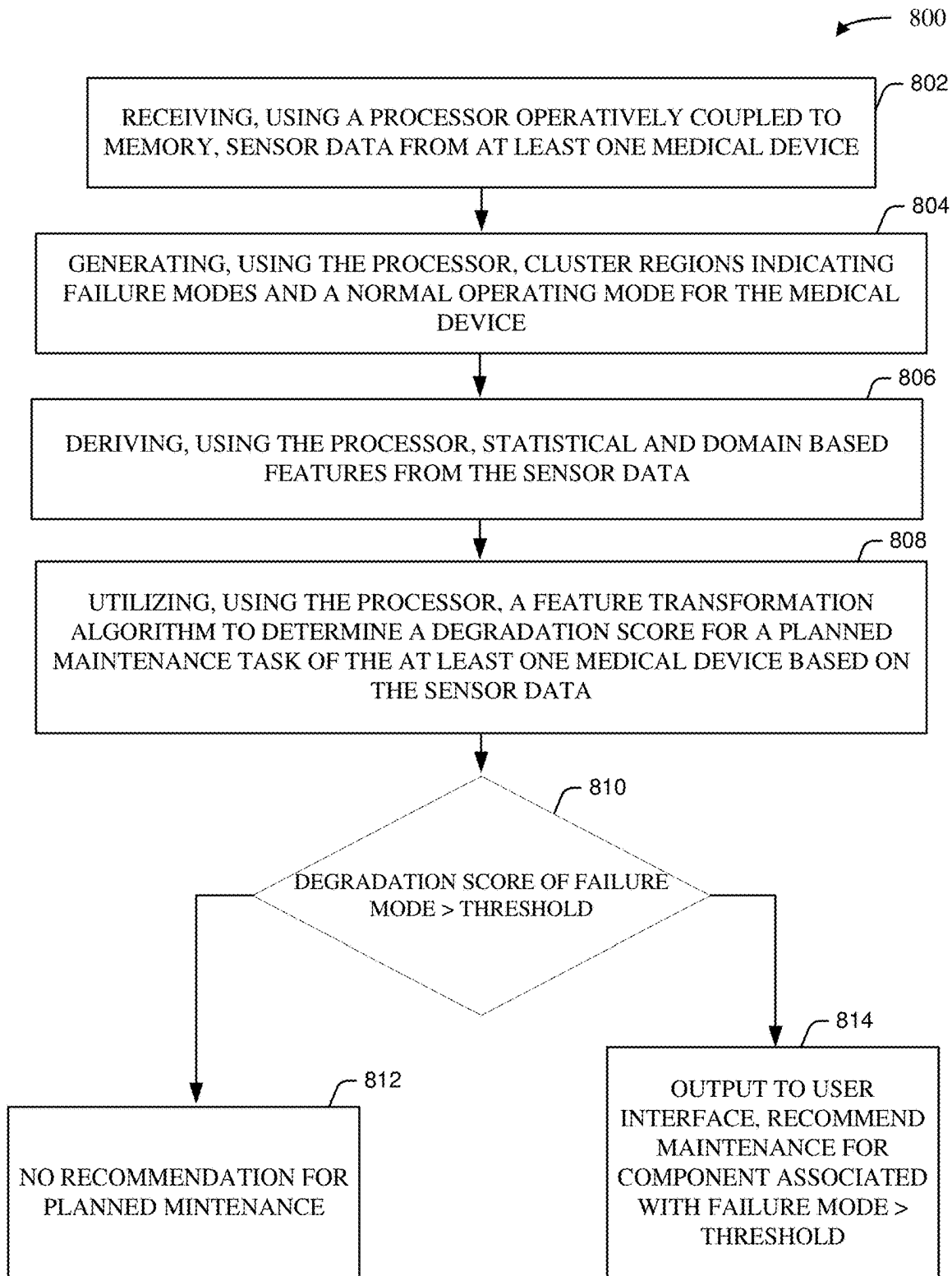


FIG. 8

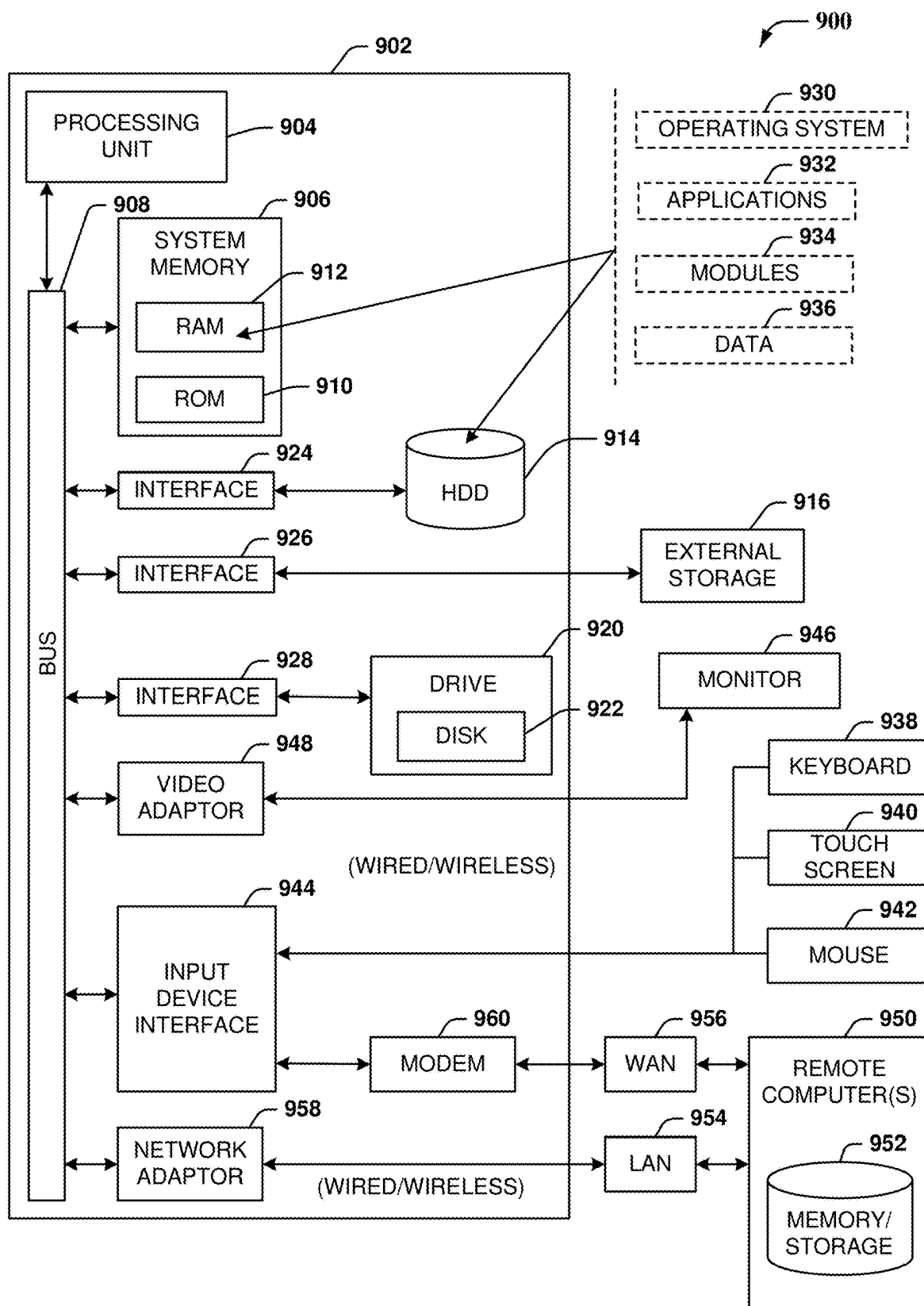


FIG.9

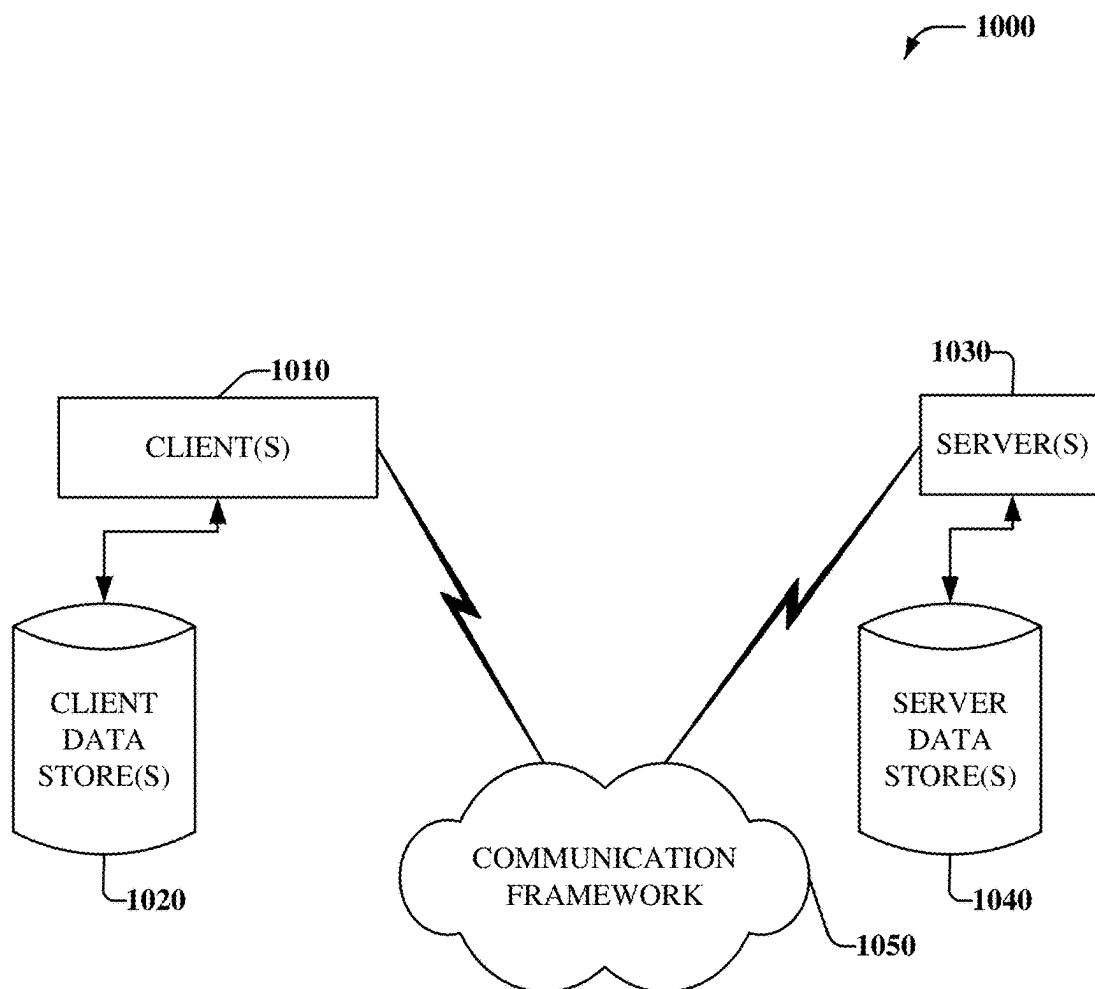


FIG. 10

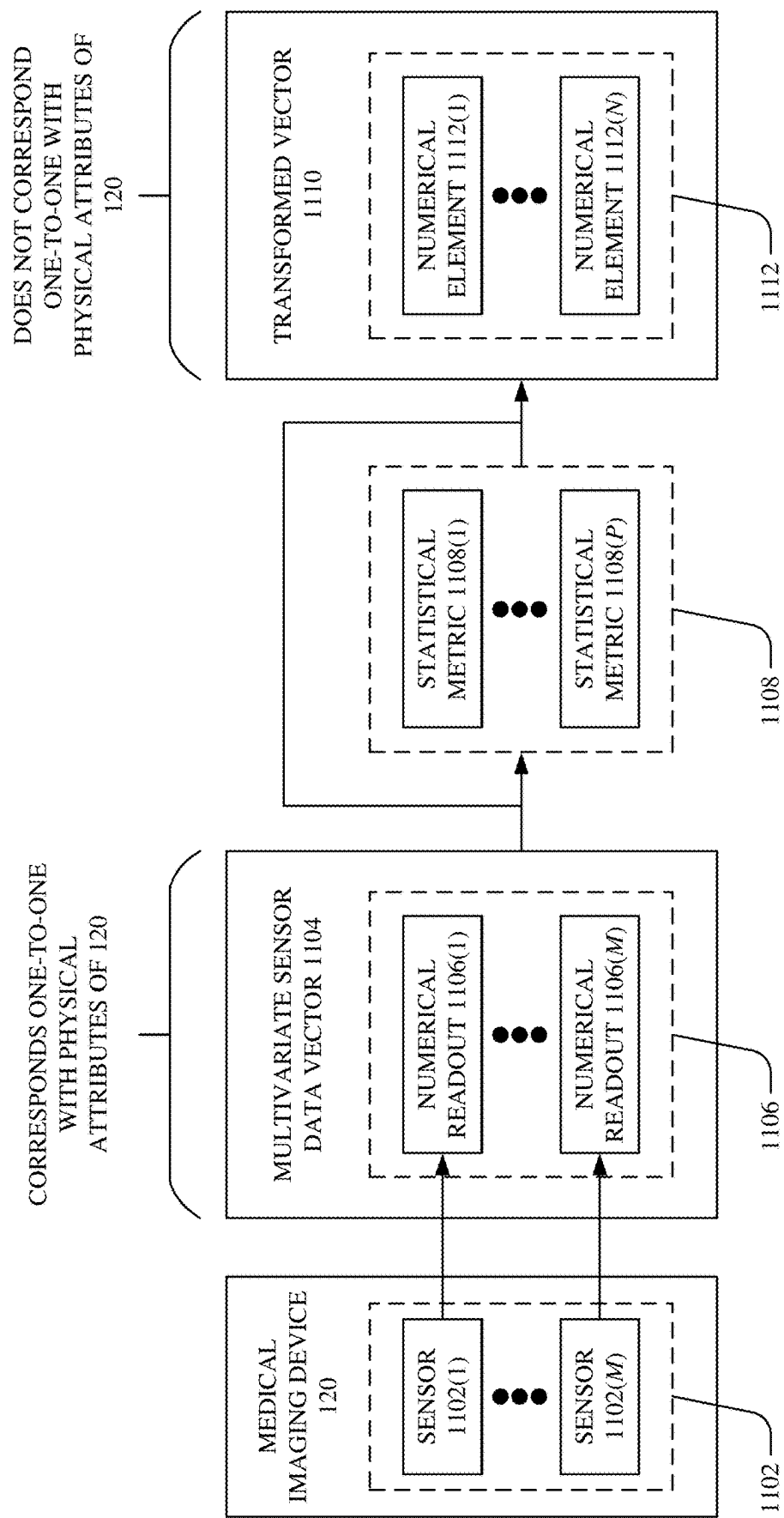


FIG. 11

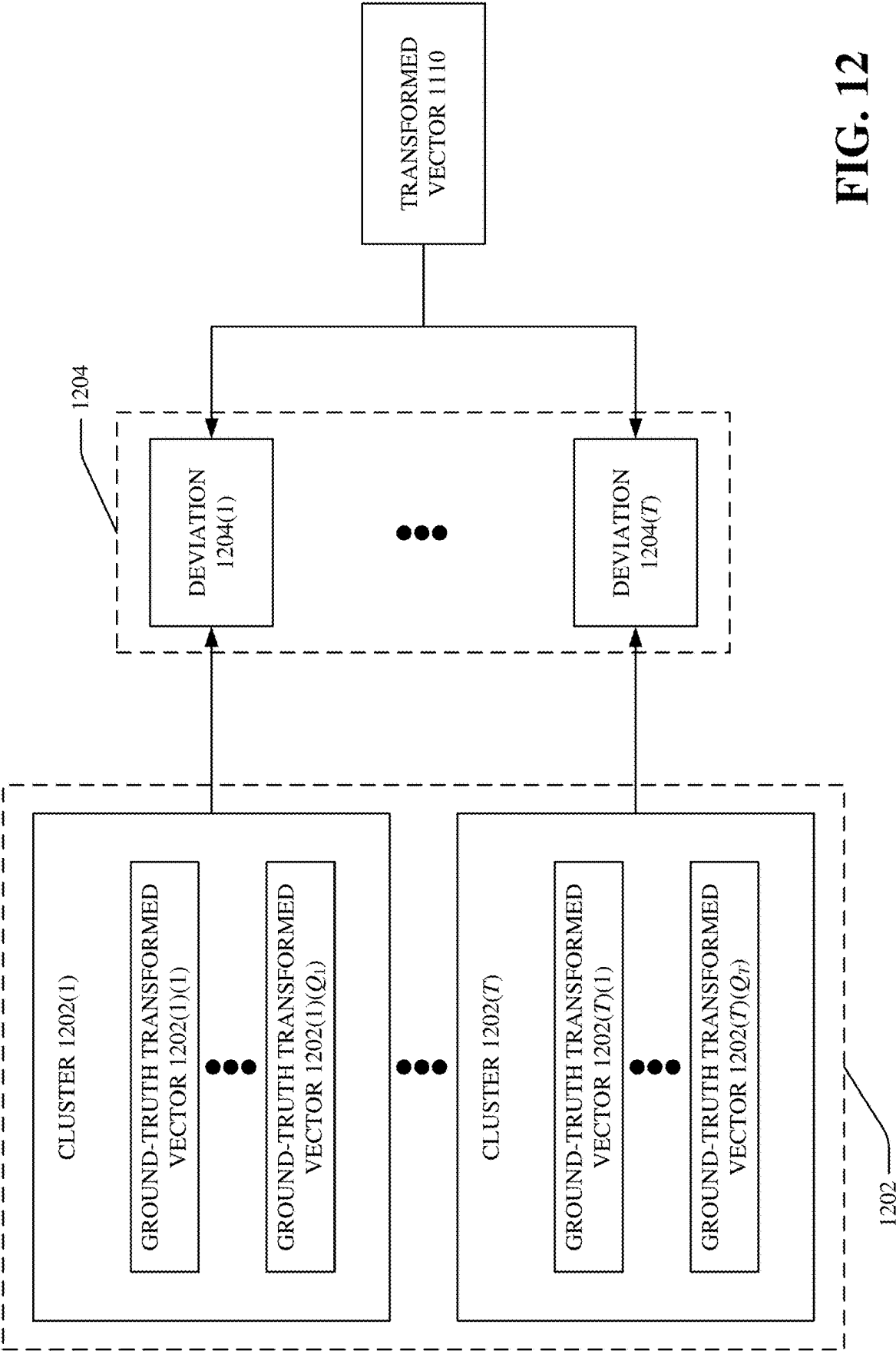


FIG. 12

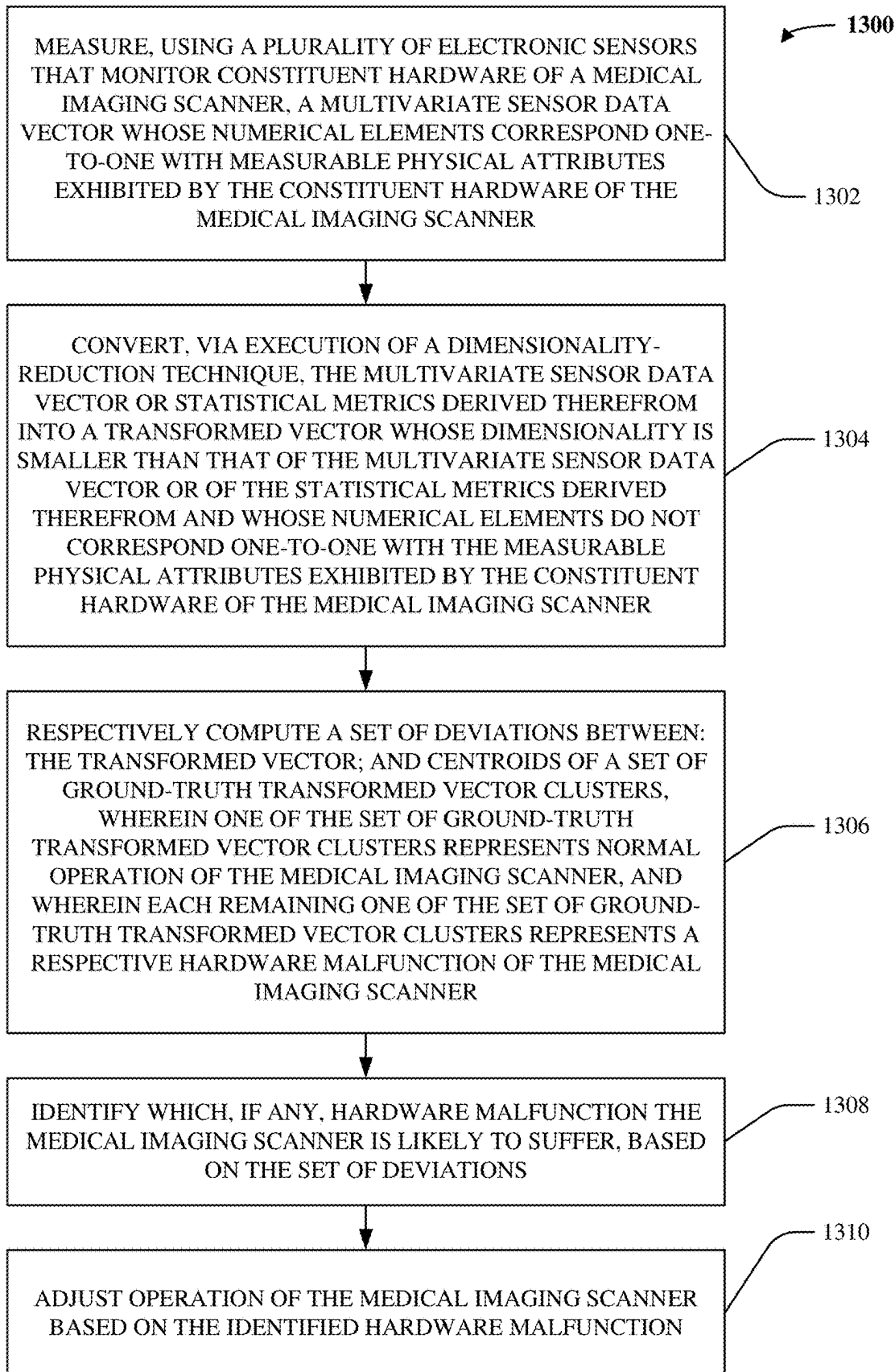


FIG. 13

SYSTEM AND METHODS FOR ENABLING REMOTE PLANNED MAINTENANCE OF MEDICAL IMAGING DEVICES

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a Continuation-In-Part of and claims priority to pending U.S. patent application Ser. No. 18/057,347, filed Nov. 21, 2022, and entitled “SYSTEM AND METHODS FOR ENABLING REMOTE PLANNED MAINTENANCE OF MEDICAL IMAGING DEVICES,” the entirety of which is hereby incorporated by reference herein.

TECHNICAL FIELD

[0002] The disclosed subject matter relates to applications of algorithms in the planned maintenance of medical imaging devices and, more particularly, the application of algorithms to enabling remote planned maintenance of medical imaging devices.

BACKGROUND

[0003] There is a growing interest in the use of algorithms and machine learning to improve the cost of maintenance of medical image devices. Conventional practice is to perform periodic planned maintenance for a system or device multiple times during a year, typically performed by a service engineer. Each planned maintenance instance for a system or device can take up to eight hours and can be executed even if a repair is not needed for the particular system or device. Further, there is a desire to reduce machine downtime and labor costs associated with the maintenance of medical imaging devices.

[0004] The above-described background relating to the remote planned maintenance of medical imaging devices is merely intended to provide a contextual overview of some current issues and is not intended to be exhaustive. Other contextual information may become further apparent upon review of the following detailed description.

SUMMARY

[0005] The following presents a summary to provide a basic understanding of one or more embodiments described herein. This summary is not intended to identify key or critical elements or delineate any scope of the particular embodiments or any scope of the claims. The sole purpose of the summary is to present concepts in a simplified form as a prelude to the more detailed description that is presented later. In one or more embodiments described herein, systems, devices, computer-implemented methods, and/or computer program products that facilitate enabling remote planned maintenance of medical imaging devices are described.

[0006] According to an embodiment, a system can comprise a memory that stores computer executable components and a processor that executes the computer executable components stored in the memory. The computer executable components can include an input component, a degradation component and a planned maintenance component. The input component can receive sensor data from at least one medical device. Additionally, the degradation component can utilize a feature transformation algorithm to determine a degradation score for a planned maintenance task of the at

least one medical device based on the sensor data. The planned maintenance component can recommend maintenance for the at least one medical device based on the degradation score.

[0007] According to another embodiment, a computer-implemented method of enabling remote planned maintenance of medical imaging devices, can comprise receiving, using a processor operatively coupled to memory, sensor data from at least one medical device. Additionally, the computer-implemented method can include utilizing, using the processor, a feature transformation algorithm to determine a degradation score for a planned maintenance task of the at least one medical device based on the sensor data. Further, the computer-implemented method can include recommending, using the processor, maintenance for the at least one medical device based on the degradation score.

[0008] According to yet another embodiment, a computer program product for enabling remote planned maintenance of medical imaging devices, the computer program product can comprise a computer readable storage medium having program instructions embodied therewith, the program instructions executable by a processor can cause the processor to receive sensor data from at least one medical device. Additionally, the computer program product can utilize, using the processor, a feature transformation algorithm to determine a degradation score for a planned maintenance task of the at least one medical device based on the sensor data. Further, the computer program product can recommend, using the processor, maintenance for the at least one medical device based on the degradation score.

DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 illustrates a block diagram of an example, non-limiting system that can facilitate evaluating the sensitivity of a tree-based model, in accordance with one or more embodiments described herein.

[0010] FIG. 2 illustrates a block diagram of an example, non-limiting system that can facilitate enabling remote planned maintenance of a medical imaging device, in accordance with one or more embodiments described herein.

[0011] FIG. 3 illustrates a block diagram of an example, non-limiting system that can facilitate enabling remote planned maintenance of a medical imaging device by performing principal component analysis via the degradation component, in accordance with one or more embodiments described herein.

[0012] FIG. 4A illustrates a 3D cluster graph of an example, non-limiting normal operating region and associated failure regions for a medical imaging device, in accordance with one or more embodiments described herein.

[0013] FIGS. 4B and 4C illustrate 2D cluster graphs of an example, non-limiting normal operating region and associated failure regions for a medical imaging device, in accordance with one or more embodiments described herein.

[0014] FIG. 5A illustrates a 3D cluster graph of an example, non-limiting normal operating region and associated failure regions for a medical imaging device via Kernel PCA, in accordance with one or more embodiments described herein.

[0015] FIG. 5B illustrates 2D cluster graphs of an example, non-limiting normal operating region and associated failure regions for a medical imaging device via Kernel PCA, in accordance with one or more embodiments described herein.

[0016] FIGS. 5C and 5D illustrate 2D cluster graphs and associated statistical graphs of an example, non-limiting system that can facilitate enabling remote planned maintenance of a medical imaging device, in accordance with one or more embodiments described herein.

[0017] FIG. 6A illustrates a graph of an example, non-limiting normal operating region and associated failure region for a medical imaging device, in accordance with one or more embodiments described herein.

[0018] FIG. 6B illustrates a degradation plot and associated component plots indicating the performance of the medical imaging device relative to normal operation, in accordance with one or more embodiments described herein.

[0019] FIG. 7 illustrates degradation plots indicating the performance of the medical imaging device relative to normal operation, in accordance with one or more embodiments described herein.

[0020] FIG. 8 illustrates a flowchart of an example, non-limiting system that can facilitate enabling remote planned maintenance of a medical imaging device, in accordance with one or more embodiments described herein.

[0021] FIG. 9 illustrates an example, non-limiting computing environment in which one or more embodiments described here can be implemented.

[0022] FIG. 10 illustrates an example, non-limiting network environment in which one or more embodiments described herein can be implemented.

[0023] FIGS. 11-13 illustrate example, non-limiting diagrams that highlight how various embodiments described herein are distinct from and better than existing techniques for facilitating predictive maintenance of medical imaging scanners.

DETAILED DESCRIPTION

[0024] The subject disclosure is now described with reference to the drawings, where like reference numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the subject disclosure. It may be evident, however, that the subject disclosure may be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to facilitate describing the subject disclosure. Furthermore, there is no intention to be bound by any expressed or implied information presented in the preceding Background or Summary sections, or in the Detailed Description section.

[0025] As alluded to above, planned maintenance can be improved in one or more various ways, and various embodiments are described herein to this end and/or other ends.

[0026] In some cases, the task of scheduling and performing planned maintenance for a medical imaging device can include inspecting the system and/or conducting repairs multiple times over a year (e.g., once a year or once every two years), and in some cases, where maintenance may not be required. Generally, maintenance activities can be scheduled at specified intervals (e.g., every 3 months) to care and service non-failed Systems/Subsystems/Components (SSCs) for the purpose of maintaining the medical device in satisfactory operating condition before a failure results in downtime. Due to the complexity of medical devices, planned maintenance can include numerous tasks (e.g., upwards of 100 in some cases) and thus, the downtime of the medical device can be high. At times, maintenance tasks can

be performed irrespective of whether a repair or upkeep is actually needed, such systems fail to consider the current state of the medical device (or component included therein). In examples, a medical device can be scheduled for maintenance, but the medical device may not actually require repair/maintenance. In one or more of a variety of manners, the device can perform longer than the expected performance duration for the device. Such repairs on devices performing to standard are wasteful and costly. Conversely, a repair can be required for a medical device after performance of the medical device degrades to a level of unacceptable performance.

[0027] Nonetheless, even though maintenance can be scheduled and performed for medical devices, a problem associated with planned maintenance, is that it can be conducted when a medical device is performing as expected, or after the medical device has degraded below an acceptable level of performance. One or more embodiments described herein can be implemented to provide service engineers the ability to decide whether a scheduled planned maintenance task is really desired or not based on the current state of the SSCs; thus enabling the current practice of on-site planned maintenance to remote planned maintenance.

[0028] Given these problems, one or more embodiments described herein can be implemented to produce a solution to one or more of these problems in the form of systems, computer-implemented methods, and/or computer program products that can facilitate the following processes: i) receiving, using a processor operatively coupled to memory, sensor data from at least one medical device; ii) utilizing, using the processor, a feature transformation algorithm to determine a degradation score for a planned maintenance task of the at least one medical device based on the sensor data; and iii) recommending, using the processor, maintenance for the at least one medical device based on the degradation score. That is, embodiments described herein include one or more systems, computer implemented methods, apparatuses and/or computer program products that can facilitate one or more of the aforementioned processes.

[0029] FIG. 1 illustrates a block diagram of an example, non-limiting system 102 that can facilitate enabling remote planned maintenance of a medical imaging device. The non-limiting system 102 can comprise one or more of a variety of components, such as memory 104, processor 106, bus 108, input component 110, degradation component 112, planned maintenance component 114, and one or more medical imaging devices 120 (e.g., which can include one or more varieties of medical devices). Additionally, the input component 110 can receive sensor data from at least one medical imaging device 120. In various embodiments, one or more of memory 104, processor 106, bus 108, input component 110, degradation component 112, planned maintenance component 114, and one or more medical imaging devices 120 can be communicatively or operatively coupled (e.g., over a bus or wireless network) to one another to perform one or more functions of the system 102.

[0030] With embodiments, the various components of the remote planned maintenance system 102 can be connected such that the degradation component 112 can utilize a feature transformation algorithm 210 to determine a degradation score for a planned maintenance task of the at least one medical imaging device 120 based on the received sensor data (e.g., via the input component 110). Repetitive

description of like elements employed in other embodiments described herein is omitted for sake of brevity. Aspects of systems (e.g., the remote planned maintenance system **102**, and the like), apparatuses or processes in various embodiments of the present invention can constitute one or more machine-executable components embodied within one or more machines (e.g., embodied in one or more computer readable mediums (or media) associated with one or more machines). Such components, when executed by the one or more machines (e.g., computers, computing devices, virtual machines, a combination thereof, and/or the like) can cause the machines to perform the operations described.

[0031] Additional description of functionalities will be further described below with reference to the example embodiments of FIGS. **1** and **2**, where repetitive description of like elements and/or processes employed in respective embodiments is omitted for sake of brevity. The remote planned maintenance system **102** can facilitate: i) receiving, using a processor operatively coupled to memory, sensor data from at least one medical imaging device **120**; ii) utilizing, using the processor, a feature transformation algorithm **210** to determine a degradation score for a planned maintenance task of the at least one medical imaging device **120** based on the sensor data; and iii) recommending, using the processor, maintenance for the at least one medical imaging device **120** based on the degradation score. The degradation score can further indicate a propensity for a particular failure mode of the at least one medical imaging device **120**. Further, the planned maintenance component **114** can include a failure threshold for the planned maintenance task associated with the medical imaging device **120**. One or more various sources of information can be utilized in determining the degradation index and/or the failure threshold for the respective medical imaging device **120**. Such information can include sensor data obtained directly from the medical imaging device **120** (e.g., raw sensor data) and/or can include performance/part information (e.g., schematic-based information from the manufacturer of the medical imaging device **120**) obtained from a linked medical imaging device **120** informational database, which can be associated with operating environment **900** (FIG. **9**).

[0032] In embodiments, such as generally illustrated in FIG. **1**, the input component **110** can be configured to receive information from one or more medical devices **120**. Further, the input component **110** can receive sensor data **202** from one or more medical imaging devices **120**. Sensor data **202** can include, but not limited to, operational data and ground truth data for the system **102** and connected medical imaging devices **120**. In examples, medical imaging devices can include, but are not limited to, medical imaging devices such as, computed tomography (CT) devices, magnetic resonance imaging (MRI) devices, positron emission tomography (PET) devices, or any other device involved in the medical imaging process. Further, the input component **110** can receive sensor data from the one or more various connected medical imaging devices **120**. For example and without limitation, sensor data **202** can include any variety of sensor parameter **122** including: raw sensor data, fan speed data, component temperature data, material degradation data, usage data, environmental data, performance data, or any other measurement that can indicate performance of the device and/or whether a repair should be conducted (e.g., up to M parameters).

[0033] In embodiments, such as where the medical imaging device **120** can be a gantry heater, sensor data **202** can include a heater outlet temperature, a first front fan speed, a second front fan speed, a high plenum temperature, and/or a low plenum temperature. For example, a normal operating region for a gantry heater can include a temperature between 26 and 24 degrees Celsius. Temperatures detected outside the normal operating region can indicate that the medical imaging device **120** should undergo maintenance or repair (such as to maintain normal operations). In embodiments, temperature values for the plenum that are lower than the acceptable threshold along with a low heater outlet temperature can correspond to a heater failure issue of the medical imaging device **120**. Low or high fan speeds can be associated with one or more conditions for repair. For example and without limitation, a high first front fan speed and a high second front fan speed can correspond to a filter clogging issue of the medical imaging device **120**. In other examples, a low first front fan speed and a low second front fan speed can correspond to one or more fan failures of the medical imaging device **120**. One or more of a variety of other parameters correlating to a failure mode for one or more components of the medical imaging device **120** can be monitored in a similar manner for one or more failure modes.

[0034] With embodiments, such as generally illustrated in FIGS. **1** and **2**, the non-limiting remote planned maintenance system **102** can utilize the sensor data **202** in combination with other data via a pre-processing component **204**. The pre-processing component **204** can be operatively connected with the processor **106**, the input component **110**, the degradation component **112**, and/or the planned maintenance component **114**. In examples, the pre-processing component **204** can include logging one or more various sensor parameters (corresponding to the medical imaging devices **120**) across tables in a data lake. Further, the non-limiting remote planned maintenance system **102** can fetch data through querying over time for the medical imaging devices **120**. For instance, sensor data **202** can include data aggregation and merging one or more model's granularity **206**. Sensor data **202** can include time series sensor data. Further, sensor data **202** can be collected/monitored over varying time frequencies and can be monitored over varying fixed time amounts (e.g., 4 hours, 8 hours, or 12 hours) and analyzed by the system **102**. The sensor data **202** can be aggregated by the pre-processing component **204** such that the various tables of the table lake include a uniform time frequency for the various sensor parameters **122**.

[0035] Such as generally illustrated in FIG. **2**, the remote planned maintenance system **102** can integrate the various medical imaging device **120** sensor information via the pre-processing component **204**. Additionally, the pre-processing component **204** can include a feature creation module **208** that can consider further sensor information. In examples, the feature creation module **208** can extract an additional set of features from the high frequency raw parameters (e.g., slope, moving average, moving standard deviation, time series decomposition using machine learning, frequency decomposition via wavelet-based techniques, etc.). The pre-processing component **204** can integrate the sensor data **202** with the additional set of features extracted

via the feature creation module 208 to facilitate predicting and determining the planned maintenance of medical imaging devices 120.

[0036] Further, the feature creation module 208 can derive statistical and domain-based features from the sensor data 202. With examples, the feature creation module can determine mean, standard deviation, median, maximum, minimum, maximum to median ratio, min to median ratio, log transformation, polynomial transformations, and various other statistical operations. The feature creation module 208 can additionally consider physics-based features for the medical imaging devices 120. In examples, the feature creation module 208 can consider the rate of change of a sensor parameter with time, as well as the instant differences of the parameters (e.g., temperature, fan speed changes, performance measurements, etc.). Additionally, the feature creation module 208 can utilize a machine learning algorithm (e.g., decision tree, random forest, etc.) for the selection of features to be monitored.

[0037] In embodiments, the remote planned maintenance system 102 (e.g., or the degradation component 112) can utilize a feature transformation algorithm 210 (e.g., an objective driven feature transformation algorithm) to further evaluate maintenance/performance of the medical imaging devices 120. The feature transformation algorithm 210 can transform the selected features of the medical imaging device 120 from a high dimensional feature space to a low dimensional feature space (e.g., where feature space can be a collection of features of the medical imaging device 120 that are further used to characterize data). Transformation can be achieved in one or more of a variety of algorithms/analysis, such as via Principal Component Analysis (e.g., and variations like Kernel Principal Component Analysis, Linear Discriminant Analysis, Facto Analysis, etc.). The resulting transformed features can enable the separation of a normal operating region for the medical imaging device 120 from a region of failure or degradation for the medical imaging device 120. The normal operating region and the failure regions for the medical imaging device 120 can be monitored so that the remote planned maintenance system 102 can determine whether maintenance should be performed on the medical imaging device 120 (e.g., or to postpone scheduled maintenance based on the acquired sensor data 202). The feature creation module 208 can utilize the above statistical information in determining the normal operating region and the failure/degradation regions for the medical imaging device 120. Further, the mean, standard deviation, and median (e.g., among other statistical operations) can be used by the feature transformation algorithm 210 to monitor and/or predict the performance of the medical imaging device 120 (such as generally illustrated in FIG. 5C).

[0038] With embodiments, such as generally illustrated in FIG. 3, the degradation component 112 can generate a new set of variables that are uncorrelated in the transformed space (e.g., via the feature transformation algorithm 210). Further, boundaries for the normal operating region and failure regions of the medical imaging device 120 can be identified as shown by plot 310. For a medical imaging device 120 operating without abnormalities, it is expected to have the feature space reside within the defined operating region (e.g., normal operating region). Data points that deviate away from the normal operating region (e.g., which can be determined via statistical and physics-based features

of the feature creation module 208) can indicate an emerging degradation in the medical imaging device 120. The normal operating region for a medical imaging device 120 can vary from device to device. Further, the normal operating region can be a single continuous region or can be one or more various disconnected regions based on the operation of the medical imaging device 120. In embodiments where regions of failure modes overlap, recommendations can be given to perform the planned maintenance tasks associated with each of the failure modes.

[0039] With embodiments, the non-limiting system 102 can separate fault data from normal operating data via principal component analysis 304 and applying a clustering algorithm 306. The degradation component 112 can further utilize machine data from existing features 302 and can employ principal component analysis 304 on the medical imaging device 120 to determine components of the medical imaging device 120 to be monitored or used in calculating a degradation of a medical imaging device 120. The degradation component 112 can divide the sensor data (e.g., and other calculated data) into one or more clusters.

[0040] For example and without limitation, such as generally illustrated by FIG. 4A, the degradation component 112 can divide the sensor data 202 into a normal operating region 402, a first failure mode 404, a second failure mode 406, and a third failure mode 408. With embodiments, the first failure mode 404 can correspond to a heater failure (e.g., where the plenum temperature can be outside the acceptable range and the heater outlet temperature can be low), the second failure mode 406 can correspond to a filter clogging issue (e.g., where the first front fan speed and the second front fan speed can be high), and the third failure mode 408 can correspond to a fan failure of the medical imaging device 120 (e.g., where the first front fan speed and the second front fan speed can be low). The normal operating region 402 can correspond to a medical imaging device 120 functioning as intended (e.g., thereby not experiencing diminished performance). The failure modes 404-408 can correspond to specific planned maintenance tasks for the medical imaging device 120 (e.g., cleaning or replacing a fan/battery).

[0041] Such as illustrated in FIGS. 4A and 4B, applying the principal component analysis 304 to the medical imaging device 120 can result in generating clusters for each of the component failures. In examples, for the failure modes, a boundary condition can be identified that defines the region of failure. The failure mode can be isolated in the transformed space and the current state of the medical imaging device 120 can be predicted based on the location of the datapoint in the associated plot 310. Boundaries of the clusters can be defined through algorithms such as K-means clustering, fuzzy clustering, density based clustering, and spectral clustering.

[0042] With embodiments, such as shown by FIGS. 4A and 4B, plot 400 represents a 3D clustering plot of principal components of the medical imaging device 120, as performed by the principal component analysis algorithm 304 and the clustering algorithm 306. The degradation component 112 can further generate one or more 2D clustering plots representative of the one or more failure modes for the associated medical imaging device 120 for further processing/analysis. 3D clustering plot 400 can be separated into a first 2D clustering plot 400', a second 2D clustering plot 400'', and a third 2D clustering plot 400'''. The first 2D

clustering plot **400'** can be utilized to determine whether the medical imaging device **120** is experiencing a heater failure. Such first 2D clustering plot **400'** can be generated to analyze performance data associated with the plenum temperatures and the heater outlet temperature (e.g., to detect plenum temperatures outside of the acceptable range and a heater outlet temperature lower than the desired threshold indicating sub-optimal performance). The second 2D clustering plot **400''** can be utilized to determine whether the medical imaging device **120** is experiencing filter clogging issues. Such second 2D clustering plot **400''** can be generated to analyze performance data associated with the first front fan speed and the second front fan speed (e.g., to detect high fan speeds greater than the desired threshold indicating sub-optimal performance). The third 2D clustering plot **400'''** can be utilized to determine whether the medical imaging device **120** is experiencing fan failures. Such third 2D clustering plot **400'''** can be generated to analyze performance data associated with the first front fan speed and the second front fan speed (e.g., to detect low fan speeds lower than the desired threshold indicating sub-optimal performance).

[0043] In embodiments, the degradation component **112** can be configured to detect an emerging failure state. The degradation component **112** can predict the type of failure based on the location of the datapoint in the transformed space. For instance, datapoints approaching the boundaries of the normal operating region can indicate to the system **102** that an emerging failure state is likely and to schedule maintenance of the medical imaging device **120** (e.g., indicating a high propensity for failure, repair, or maintenance).

[0044] With embodiments, for datapoints lying outside the normal operating region **402**, the deviation can be quantified through a degradation index. Further, the deviation can be calculated via a variety of methods (e.g., Euclidean distance, the projection of each datapoint from the center of the normal operating region, etc.). The quantified deviation can be scaled between a range of 0-100 to indicate the severity of the degradation for the associated failure. The degradation component can derive a threshold for the failure mode degradation indexes where components of the medical imaging device **120** are likely to fail (e.g., have a propensity to fail). For the failure modes **404-408**, a degradation index **404'**, **406'**, **408'** (e.g., such as seen in FIG. 3) can be calculated to analyze the degradation state of the medical imaging device **120**. A higher degradation index can indicate a higher failure severity (where the medical imaging device **120** is more sensitive to failure and/or failure is more likely to result).

[0045] Turning now to FIG. 4C, the first 2D clustering plot **400'** can be used to determine whether there is a heater failure of the medical imaging device **120**; and further, whether maintenance should be performed. The failure modes **404-408** (e.g., the associated 2D clustering plots) can include a fault direction **410**. The fault direction **410** can indicate that one or more components of the medical imaging device **120** includes performance data indicative of failure. For example and without limitation, the fault direction **410** includes values for components that lie outside the normal operating region **402**. The planned maintenance of a medical imaging device **120** can be scheduled for components as performance values for such components of the medical imaging device **120** move away from a direction of nominal data **412**.

[0046] Such as generally illustrated in FIGS. 5A and 5B, the objective driven feature transformation algorithm **210** can employ Kernel Principal Component Analysis (PCA) **304** to generate a 3D clustering plot **500** of the principal components of the medical imaging device **120**. The objective driven feature transformation algorithm **210** (e.g., and/or the degradation component **112**) can further generate one or more 2D clustering plots representative of one or more failure modes for further processing/analysis. 3D clustering plot **500** can be separated into a first 2D clustering plot **500'**, a second 2D clustering plot **500''**, and a third 2D clustering plot **500'''**. A first failure mode **502** can correspond to a low plenum temperature; a second failure mode **504** can correspond to a low fan speed and high outlet temperature; and a third failure mode **506** can correspond to a high plenum temperature and a high fan speed.

[0047] Turning next to FIGS. 5C and 5D, the degradation component **112** can utilize statistical and physics-based features from the feature creation module **208** of the pre-processing component **204** to determine whether the values of the components of the medical imaging device **120** are approaching the region of failure (e.g., where the component data is moving in a fault direction on the associated 2D plot). The 3D clustering plot **500** can be used to generate the first 2D clustering plot **500'**; and further, the pre-processing component **204** (e.g., comprising data aggregation and merging one or more model's granularity **206** and the feature creation module **208**) can statistically process the sensor data **202** to determine the mean of the low plenum temperature (shown by plot **510**) and the mean of the high plenum temperature (shown by plot **512**). In a similar manner, the pre-processing component **204** can additionally, statistically process the sensor data **202** to determine the mean of the first front fan speed (shown by plot **514**) and the mean of the second front fan speed (shown by plot **516**). As illustrated in FIG. 5D, the pre-processing component **204** can statistically process the sensor data **202** to determine the mean of the high plenum temperature (plot **518**), the mean of the low plenum temperature (plot **520**), the mean of the first front fan speed (plot **522**), and the mean of the second front fan speed (plot **524**).

[0048] In embodiments, the degradation component **112** can derive a threshold for each failure mode based on statistical or machine learning approaches (e.g., as executed by the pre-processing component **204**). As the degradation index **404'-408'** crosses the failure threshold of failure modes **404-408**, the planned maintenance component **114** can generate a recommendation to a service engineer (e.g., remotely via a user interface **720**) to perform the planned maintenance task associated with the failure modes **404-408** (see e.g., FIG. 4). Further, for overlapping regions of failure modes **404-408**, where the degradation index is within the threshold, the service engineer can be asked to skip the associated planned maintenance task associated with the failure mode **404-408**.

[0049] Such as generally illustrated in FIG. 6A, the degradation component **112** can measure a degree of separation between the normal operating region **602** and the degradation/failure mode region **604** by converting the separation into a degradation index (e.g., corresponding degradation indexes **404'**, **406'**, **408'** such as in FIG. 2, or degradation index **620**). The degradation index **620** can indicate the propensity for a particular failure mode (e.g., that a heater is likely to fail resulting in sub-optimal of the medical imaging

device 120). Further, the degradation index 620 includes degradation scores linked to specific planned maintenance tasks, such as replacing a gantry heater. The degradation component 112 can utilize a root cause algorithm to determine the necessary repair based on the acquired sensor data 202 in combination with the degradation index 620. The root cause algorithm can be integrated with one or more repair manuals, reference manuals, root cause tables, and/or one or more maintenance algorithms to determine the necessary repair so that the medical imaging device 120 can return to optimal performance. The root cause algorithm can include information for a wide range of medical imaging devices 120 and components thereof to accurately diagnose and assess the root cause for the medical imaging device 120 failure.

[0050] Further, with embodiments, the degradation index 620 can evaluate information from the high plenum temperature (plot 622) and the low plenum temperature (plot 624) during generation of the index. Such as can be seen from FIG. 6B, plot of degradation index 620 indicates a high degradation score for the gantry heater at point A on degradation index 620, and plots 622, 624 that can correspond to temperatures of the low plenum temperature and the high plenum temperature outside acceptable tolerances. Further, replacement of the gantry heater occurs at point B on degradation index 620, and plots 622, 624 resulting in the low plenum temperature and the high plenum temperature returning to a range of normal operation, and the degradation score of the gantry heater failure mode returning to a normal value (e.g., corresponding to the medical imaging device 120 operating normally and having correlating performance data indicating the same). Maintenance tasks including a high degradation score (e.g., a score that exceeds an acceptable performance threshold determined by the user and operating conditions of the medical imaging device 120) can be recommended to a service engineer via the user interface 720. For example, if the degradation score is greater than the desired threshold, such a score can indicate that the associated medical imaging device 120 includes a high failure sensitivity. Further, a high failure sensitivity can indicate that the medical imaging device 120 is likely to fail for one or more reasons (e.g., the medical imaging device 120 is likely to fail prior to the subsequent planned maintenance of the medical imaging device 120). In other embodiments, a high failure sensitivity can indicate that the performance of a component of the medical imaging device 120 is trending towards the fault direction for a particular failure region, thus maintenance can be required in urgent and non-urgent manners depending on how fast the component is trending towards the failure region.

[0051] In examples, the planned maintenance component 114 can process the degradation index 620 via machine learning to determine a manner of planned maintenance for the various components of the medical imaging device 120 (e.g., the various failure modes/regions). Turning next to FIG. 7, the non-limiting remote planned maintenance system 102 can evaluate one or more degradation scores associated with failure modes of the medical imaging device 120. A first degradation plot 700 and a second degradation plot 702 can be generated for a failure mode (e.g., such as gantry heater failure) of the medical imaging device 120. Further, the first degradation plot 700 can include a first reference point 704 where the degradation score for the medical imaging device 120 increases indicating improper perfor-

mance. In response, planned maintenance can be scheduled to correct performance of the medical imaging device 120 (e.g., at second reference point 706). The degradation score for heater failure can degrease and return to normalcy after replacing the heater via planned maintenance. In other embodiments, pre-scheduled maintenance can be waived if the degradation score indicates that the components of the associated medical imaging device are performing as expected. Additionally, degradation scores can return to normalcy without conducting pre-planned maintenance (such as at reference point 708), and such performance can be considered in scheduling the next planned maintenance.

[0052] With examples, in a similar manner with respect to first degradation plot 700, the second degradation plot 702 can include a first reference point 710 indicating scheduled and conducted planned maintenance in response to detecting high degradation scores. As can be seen from the second degradation plot 702, the degradation score for heater failure can decrease and returns to normalcy after conducting maintenance (e.g., via cleaning and/or component replacement). Further, a second reference point 712 can indicate a high degradation score returning to normalcy without planned maintenance. Additionally, a third reference point 714 can indicate a high degradation score due to heater failure whereby maintenance has not been pre-scheduled and maintenance should be performed to return the heater to optimal performance for the medical imaging device 120.

[0053] In examples, the first degradation plot 700 and the second degradation plot 702 can be visible within a user-interface 720 (as shown in FIG. 2). Further, the non-limiting remote planned maintenance system 102 can display one or more maintenance recommendations via the user-interface 720 that can be similar in appearance to plots 700 and 702.

[0054] For example and without limitation, planned maintenance can include any combination of the following actions: (1) powering down the medical imaging device 120, (2) powering down a component of the medical imaging device 120, (3) generating/transmitting an alert that the medical imaging device 120 is not functioning properly, (4) reducing performance of a component of the medical imaging device 120; (5) disconnecting the medical imaging device 120; (5) replacing a component (e.g., heat sink, fan, cooling fluid, bearings, and/or other properties of the components of the medical imaging device 120; and (6) replacing the medical imaging device 120. If maintenance has been scheduled for a medical imaging device 120, the remote planned maintenance system 102 can selectively disconnect components of the medical imaging device 120 and/or the medical imaging device 120 in whole to facilitate a safe and efficient repair process. For example, the remote planned maintenance system 102 can disconnect power to the medical imaging device 120 and await maintenance/repair. In addition, the remote planned maintenance system 102 can generate one or more alerts that the medical imaging device 120 will disconnect and/or will be inoperable until maintenance can be conducted (which can be remotely scheduled).

[0055] The user-interface 720 can be a checklist or a report that includes visuals (e.g., supporting charts, graphs, data, etc.) for service and/or field engineers to consider when conducting maintenance. The user interface 720 can additionally include a description of the current state of the components corresponding with the identified failure modes for the medical imaging device 120 (e.g., which can include

highlights to direct service engineers to important information on the user interface **720**). For failure modes where the degradation index is high, the user interface **720** can display a root cause for the high degradation index. The user planned maintenance component **114** can employ a root cause algorithm in identifying the root cause for a failure mode, thus integrating the degradation index. The user interface **720** can be connected on site with the various medical imaging devices **120** such that a service engineer can easily access and interact with the user interface **720**.

[0056] In embodiments, the non-limiting remote planned maintenance system **102** can be evaluated by one or more various metrics. For example and without limitation, such metrics can include a recall percentage and a percentage of planned maintenance tasks avoided. The recall percentage can be expressed as:

$$\frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}.$$

The recall percentage can reflect the percentage of failures/errors/issues that are detected of the medical imaging device **120**. Further, false negatives can represent situations where the remote planned maintenance system **102** misses prediction of a defective component for the associated medical imaging device **120**. True positives can represent situations where the component is defective and the remote planned maintenance system **102** has accurately predicted and/or diagnosed such failure. Additionally, the percentage of planned maintenance tasks avoided can be expressed as:

$$\frac{\text{Number of planned maintenance tasks avoided}}{\text{Total planned maintenance tasks}}.$$

The precision of the non-limiting remote planned maintenance system **102** can assessed and expressed as

$$\frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}.$$

(e.g., where false positives can be instances where a failure/error/issue is detected, however does not actually occur). An increased precision of the remote planned maintenance system **102** can be associated with reduced false calls (failure calls) for the medical imaging device **120**. An increased recall percentage of the remote planned maintenance system **102** can be associated with an increased rate of capture of failures for the medical imaging device **120**.

[0057] With embodiments, a computer-implemented method of remote planned maintenance **800** can comprise a first step of receiving, using a processor operatively coupled to memory, sensor data **202** from at least one medical imaging device **120** (step **802**). The computer implemented method can further include generating, using the processor, cluster regions indicating failure modes and a normal operating mode for the medical imaging device **120** (step **804**). In generating the cluster regions indicating failure and normal operating modes, the method can include the pre-processing component **204** (e.g., the feature creation module **808**) deriving statistical and domain-based features from the

sensor data **202** (step **806**). Further, the feature creation module can determine statistical operations of the sensor data **202** for processing the operating regions for the medical imaging device **120**. In such a step, the feature creation module **208** can identify the components that contribute to successful performance of the medical device **120** and transmit the identified components to the objective driven feature transformation algorithm **210**.

[0058] Additionally, the computer implemented method **800** can include utilizing, using the processor, a feature transformation algorithm **210** to determine a degradation score for a planned maintenance task of the at least one medical imaging device **120** based on the sensor data **202** (step **808**). Further, the feature transformation algorithm **210** can be utilized to transform the selected features of the medical imaging device **120** from a high dimensional feature space to a low dimensional feature space (e.g., for more efficient processing and/or data analysis). The feature transformation algorithm **210** can transform the data such that the degradation component **112** can determine the performance of the medical imaging device **120** by comparing historical and current operation of the components with the normal operating region and the failure operating regions. Statistical operations utilized by the feature transformation algorithm **210** can be used to monitor and/or predict the performance of the medical imaging device **120** with respect to the normal operating region and failure regions (that are to be determined by the method **800**).

[0059] In embodiments, the computer-implemented method of remote planned maintenance **800** can include determining thresholds for components for the failure modes/regions of the medical imaging device **120**. Thresholds for failure modes/regions can be determined by applying the clustering algorithm **306**. For example and without limitation, the method **800** can include determining thresholds for failure modes of the medical imaging device **120** by generating a 3D clustering plot via the principal component analysis algorithm **304** and the clustering algorithm **306**. Further, the degradation component **112** can generate one or more resulting 2D cluster plots from the 3D clustering plot reflective of the one or more failure modes/regions for the medical imaging device **120**. The degradation component can accurately assess performance and/or maintenance of the medical imaging device **120** via the one or more resulting 2D cluster plots.

[0060] With embodiments, the degradation component can determine the severity of datapoints (e.g., sensor data **202**) lying outside of the normal operating region. The severity can be determined via the deviation from the normal operating region. Further, datapoints that are farther away from the normal operating region can indicate a higher degree of sub-optimal performance (e.g., a likelihood of medical imaging device **120** failure).

[0061] The remote planned maintenance method **800** can include determining whether the degradation score of a failure mode is greater than the threshold (e.g., for repair) (step **810**). If the degradation score is not greater than the threshold, the planned maintenance component **114** can provide no recommendations for planned maintenance (step **812**). For example and without limitation, if the threshold is not exceeded, the planned maintenance component **114** can recommend not performing (e.g., skipping) the planned maintenance for the medical imaging device **120**. With embodiments, planned maintenance tasks can be checked off

a checklist (e.g., having any number of planned maintenance tasks) for the associated medical imaging device **120** by the planned maintenance component **114**. Further, tasks can be checked off a list instead of being conducted. Conversely, if the degradation score is greater than the threshold, the planned maintenance component **114** can transmit an alert to the user interface **720** and can recommend maintenance for the associated component of the failure mode corresponding to the high degradation score (e.g., a degradation score that exceeded a limit set by the user or by the functionalities of the specific medical imaging device **120**) (step **814**). In examples, the user interface **720** can include any variety of informational medium to effectively communicate an alert (e.g., graphs, tables, visualizations, data, root cause, etc.). Further, in transmitting an alert, the planned maintenance component **114** can perform one or more of a variety of actions. The planned maintenance component **114** can disconnect the medical imaging device **120** from the power source, selectively disconnect one or more components of the medical imaging device **120** from the power source, and remotely reduce performance of the medical imaging device **120** (e.g., enable a medical device to be in a lower power mode to reduce component operational demand). The planned maintenance component **114** can generate an alert while the device is isolated from further damage (e.g., disconnected from power and/or operation) and awaiting repair.

[0062] In order to provide additional context for various embodiments described herein, FIG. 9 and the following discussion are intended to provide a brief, general description of a suitable computing environment **900** in which the various embodiments of the embodiment described herein can be implemented. While the embodiments have been described above in the general context of computer-executable instructions that can run on one or more computers, those skilled in the art will recognize that the embodiments can be also implemented in combination with other program modules and/or as a combination of hardware and software.

[0063] Generally, program modules include routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the various methods can be practiced with other computer system configurations, including single-processor or multi-processor computer systems, minicomputers, mainframe computers, Internet of Things (IoT) devices, distributed computing systems, as well as personal computers, handheld computing devices, microprocessor-based or program-mable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0064] The illustrated embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0065] Computing devices typically include a variety of media, which can include computer-readable storage media, machine-readable storage media, and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media or machine-readable storage media can be any available storage media that can be accessed by the computer and includes both volatile and nonvolatile media, removable and non-

removable media. By way of example, and not limitation, computer-readable storage media or machine-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable or machine-readable instructions, program modules, structured data, or unstructured data.

[0066] Computer-readable storage media can include, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD), Blu-ray disc (BD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, solid state drives or other solid state storage devices, or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory, or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0067] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries, or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0068] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and includes any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media include wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared, and other wireless media.

[0069] With reference again to FIG. 9, the example environment **900** for implementing various embodiments of the aspects described herein includes a computer **902**, the computer **902** including a processing unit **904**, a system memory **906** and a system bus **908**. The system bus **908** couples system components including, but not limited to, the system memory **906** to the processing unit **904**. The processing unit **904** can be any of various commercially available processors. Dual microprocessors and other multi-processor architectures can also be employed as the processing unit **904**.

[0070] The system bus **908** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory **906** includes ROM **910** and RAM **912**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, where BIOS contains the basic routines that help transfer information between elements within the computer **902**, such as during startup. The RAM **912** can also include a high-speed RAM such as static RAM for caching data.

[0071] The computer 902 further includes an internal hard disk drive (HDD) 914 (e.g., EIDE, SATA), one or more external storage devices 916 (e.g., a magnetic floppy disk drive (FDD) 916, a memory stick or flash drive reader, a memory card reader, etc.) and an optical disk drive 920 (e.g., which can read or write from a CD-ROM disc, a DVD, a BD, etc.). While the internal HDD 914 is illustrated as located within the computer 902, the internal HDD 914 can also be configured for external use in a suitable chassis (not shown). Additionally, while not shown in environment 900, a solid-state drive (SSD) could be used in addition to, or in place of, an HDD 914. The HDD 914, external storage device(s) 916 and optical disk drive 920 can be connected to the system bus 908 by an HDD interface 924, an external storage interface 926 and an optical drive interface 928, respectively. The interface 924 for external drive implementations can include at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0072] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer 902, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to respective types of storage devices, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, whether presently existing or developed in the future, could also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0073] A number of program modules can be stored in the drives and RAM 912, including an operating system 930, one or more application programs 932, other program modules 934 and program data 936. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM 912. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0074] Computer 902 can optionally comprise emulation technologies. For example, a hypervisor (not shown) or other intermediary can emulate a hardware environment for operating system 930, and the emulated hardware can optionally be different from the hardware illustrated in FIG. 9. In such an embodiment, operating system 930 can comprise one virtual machine (VM) of multiple VMs hosted at computer 902. Furthermore, operating system 930 can provide runtime environments, such as the Java runtime environment or the .NET framework, for applications 932. Runtime environments are consistent execution environments that allow applications 932 to run on any operating system that includes the runtime environment. Similarly, operating system 930 can support containers, and applications 932 can be in the form of containers, which are lightweight, standalone, executable packages of software that include, e.g., code, runtime, system tools, system libraries and settings for an application.

[0075] Further, computer 902 can be enabled with a security module, such as a trusted processing module (TPM). For

instance, with a TPM, boot components hash next in time boot components, and wait for a match of results to secured values, before loading a next boot component. This process can take place at any layer in the code execution stack of computer 902, e.g., applied at the application execution level or at the operating system (OS) kernel level, thereby enabling security at any level of code execution.

[0076] A user can enter commands and information into the computer 902 through one or more wired/wireless input devices, e.g., a keyboard 938, a touch screen 940, and a pointing device, such as a mouse 942. Other input devices (not shown) can include a microphone, an infrared (IR) remote control, a radio frequency (RF) remote control, or other remote control, a joystick, a virtual reality controller and/or virtual reality headset, a game pad, a stylus pen, an image input device, e.g., camera(s), a gesture sensor input device, a vision movement sensor input device, an emotion or facial detection device, a biometric input device, e.g., fingerprint or iris scanner, or the like. These and other input devices are often connected to the processing unit 904 through an input device interface 944 that can be coupled to the system bus 908, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a USB port, an IR interface, a BLUETOOTH® interface, etc.

[0077] A monitor 946 or other type of display device can be also connected to the system bus 908 via an interface, such as a video adapter 948. In addition to the monitor 946, a computer typically includes other peripheral output devices (not shown), such as speakers, printers, etc.

[0078] The computer 902 can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) 950. The remote computer(s) 950 can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically includes many or all of the elements described relative to the computer 902, although, for purposes of brevity, only a memory/storage device 952 is illustrated. The logical connections depicted include wired/wireless connectivity to a local area network (LAN) 954 and/or larger networks, e.g., a wide area network (WAN) 956. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

[0079] When used in a LAN networking environment, the computer 902 can be connected to the local network 954 through a wired and/or wireless communication network interface or adapter 958. The adapter 958 can facilitate wired or wireless communication to the LAN 954, which can also include a wireless access point (AP) disposed thereon for communicating with the adapter 958 in a wireless mode.

[0080] When used in a WAN networking environment, the computer 902 can include a modem 960 or can be connected to a communications server on the WAN 956 via other means for establishing communications over the WAN 956, such as by way of the Internet. The modem 960, which can be internal or external and a wired or wireless device, can be connected to the system bus 908 via the input device interface 944. In a networked environment, program modules depicted relative to the computer 902 or portions

thereof, can be stored in the remote memory/storage device **952**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

[0081] When used in either a LAN or WAN networking environment, the computer **902** can access cloud storage systems or other network-based storage systems in addition to, or in place of, external storage devices **916** as described above. Generally, a connection between the computer **902** and a cloud storage system can be established over a LAN **954** or WAN **956** e.g., by the adapter **958** or modem **960**, respectively. Upon connecting the computer **902** to an associated cloud storage system, the external storage interface **926** can, with the aid of the adapter **958** and/or modem **960**, manage storage provided by the cloud storage system as it would other types of external storage. For instance, the external storage interface **926** can be configured to provide access to cloud storage sources as if those sources were physically connected to the computer **902**.

[0082] The computer **902** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, store shelf, etc.), and telephone. This can include Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0083] Referring now to FIG. **10**, there is illustrated a schematic block diagram of a computing environment **1000** in accordance with this specification. The system **1000** includes one or more client(s) **1002**, (e.g., computers, smart phones, tablets, cameras, PDA's). The client(s) **1002** can be hardware and/or software (e.g., threads, processes, computing devices). The client(s) **1002** can house cookie(s) and/or associated contextual information by employing the specification, for example.

[0084] The system **1000** also includes one or more server(s) **1004**. The server(s) **1004** can also be hardware or hardware in combination with software (e.g., threads, processes, computing devices). The servers **1004** can house threads to perform transformations of media items by employing aspects of this disclosure, for example. One possible communication between a client **1002** and a server **1004** can be in the form of a data packet adapted to be transmitted between two or more computer processes wherein data packets may include coded analyzed headspaces and/or input. The data packet can include a cookie and/or associated contextual information, for example. The system **1000** includes a communication framework **1006** (e.g., a global communication network such as the Internet) that can be employed to facilitate communications between the client(s) **1002** and the server(s) **1004**.

[0085] Communications can be facilitated via a wired (including optical fiber) and/or wireless technology. The client(s) **1002** are operatively connected to one or more client data store(s) **1008** that can be employed to store information local to the client(s) **1002** (e.g., cookie(s) and/or associated contextual information). Similarly, the server(s) **1004** are operatively connected to one or more server data store(s) **1010** that can be employed to store information local to the servers **1004**.

[0086] In one exemplary implementation, a client **1002** can transfer an encoded file, (e.g., encoded media item), to server **1004**. Server **1004** can store the file, decode the file, or transmit the file to another client **1002**. It is noted that a client **1002** can also transfer uncompressed file to a server **1004** and server **1004** can compress the file and/or transform the file in accordance with this disclosure. Likewise, server **1004** can encode information and transmit the information via communication framework **1006** to one or more clients **1002**.

[0087] The illustrated aspects of the disclosure may also be practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0088] The above description includes non-limiting examples of the various embodiments. It is, of course, not possible to describe every conceivable combination of components or methods for purposes of describing the disclosed subject matter, and one skilled in the art may recognize that further combinations and permutations of the various embodiments are possible. The disclosed subject matter is intended to embrace all such alterations, modifications, and variations that fall within the spirit and scope of the appended claims.

[0089] With regard to the various functions performed by the above-described components, devices, circuits, systems, etc., the terms (including a reference to a “means”) used to describe such components are intended to also include, unless otherwise indicated, any structure(s) which performs the specified function of the described component (e.g., a functional equivalent), even if not structurally equivalent to the disclosed structure. In addition, while a particular feature of the disclosed subject matter may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application.

[0090] The terms “exemplary” and/or “demonstrative” as used herein are intended to mean serving as an example, instance, or illustration. For the avoidance of doubt, the subject matter disclosed herein is not limited by such examples. In addition, any aspect or design described herein as “exemplary” and/or “demonstrative” is not necessarily to be construed as preferred or advantageous over other aspects or designs, nor is it meant to preclude equivalent structures and techniques known to one skilled in the art. Furthermore, to the extent that the terms “includes,” “has,” “contains,” and other similar words are used in either the detailed description or the claims, such terms are intended to be inclusive—in a manner similar to the term “comprising” as an open transition word—without precluding any additional or other elements.

[0091] The term “or” as used herein is intended to mean an inclusive “or” rather than an exclusive “or.” For example, the phrase “A or B” is intended to include instances of A, B, and both A and B. Additionally, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless either otherwise specified or clear from the context to be directed to a singular form.

[0092] The term “set” as employed herein excludes the empty set, i.e., the set with no elements therein. Thus, a “set”

in the subject disclosure includes one or more elements or entities. Likewise, the term “group” as utilized herein refers to a collection of one or more entities.

[0093] The description of illustrated embodiments of the subject disclosure as provided herein, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosed embodiments to the precise forms disclosed. While specific embodiments and examples are described herein for illustrative purposes, various modifications are possible that are considered within the scope of such embodiments and examples, as one skilled in the art can recognize. In this regard, while the subject matter has been described herein in connection with various embodiments and corresponding drawings, where applicable, it is to be understood that other similar embodiments can be used or modifications and additions can be made to the described embodiments for performing the same, similar, alternative, or substitute function of the disclosed subject matter without deviating therefrom. Therefore, the disclosed subject matter should not be limited to any single embodiment described herein, but rather should be construed in breadth and scope in accordance with the appended claims below.

[0094] To elaborate on various aspects disclosed above, consider the following discussion.

[0095] A medical imaging scanner (e.g., computed tomography (CT) scanner, magnetic resonance imaging (MRI) scanner, X-ray scanner, ultrasound scanner, positron emission tomography (PET) scanner, nuclear medicine (NM) scanner) can be considered as a complex electromechanical system that is built from, constructed with, or otherwise made up of various pieces of constituent hardware (e.g., cathodes, anodes, filaments, gantries, plenums, X-ray detectors, heat exchangers, fluid pumps, cooling fans). Such constituent hardware can experience degradation (e.g., wear and tear, fractures, corrosion, plastic deformation) over time. As the constituent hardware of the medical imaging scanner becomes progressively more degraded, the performance or operation of the medical imaging scanner can commensurately suffer (e.g., scanning a medical patient using an excessively degraded scanner might result in scanned images that are of poor quality, might result in an incomplete image acquisition that exposes the patient to ionizing radiation, contrast media or other disadvantages without resulting in useful images, might risk catastrophically damaging the scanner, or might risk exposing the medical patient to excessive or dangerous dosages of radiation). To avoid or remedy such performance problems, the medical imaging scanner can be subjected to maintenance or servicing tasks (e.g., hardware repairs, hardware replacements, hardware decontaminations, hardware calibrations) performed by technicians or field engineers.

[0096] According to many existing techniques, maintenance or servicing tasks are performed on the medical imaging scanner in a fixed-schedule fashion (e.g., X-ray tubes replaced every 12 months; gantry slip ring cleaned every 5 million gantry rotations). Unfortunately, such fixed-schedule techniques are prone to over-servicing or under-servicing the medical imaging scanner (e.g., prone to performing maintenance or servicing tasks long before they are needed, which can be excessively wasteful in terms of cost, effort, or scanner downtime; or prone to performing maintenance or servicing tasks long after they are needed, which can lead to significant spans of time during which the above-mentioned performance problems can occur).

[0097] To address this issue of over-servicing or under-servicing, other existing techniques perform maintenance or servicing tasks in a predictive fashion. That is, such other existing techniques measure physical attributes (e.g., temperatures, pressures, displacements, velocities) of the constituent hardware of the medical imaging scanner, use those measured physical attributes to proactively forecast a future time at which the medical imaging scanner will or is expected to achieve an excessive amount of degradation, and recommend or call for maintenance or servicing tasks to be performed shortly before such forecasted time. In this way, the maintenance or servicing tasks can be performed prior to the excessively degraded state being reached, thereby avoiding the above-described performance problems, but not so much prior as to wastefully or unnecessarily increase servicing costs or downtime.

[0098] Such other existing techniques fall into two categories: a threshold category; and a machine learning category.

[0099] Existing techniques belonging to the threshold category forecast the future time at which excessive degradation is expected to occur by comparing each of the measured physical attributes of the constituent hardware of the medical imaging scanner to respective threshold values (e.g., comparing a measured cooling fan speed to a known or defined speed threshold; comparing a measured heat exchanger outlet temperature to a known or defined temperature threshold; comparing a measured cathode voltage to a known or defined voltage threshold). If all of those threshold values are satisfied, it can be concluded that the medical imaging scanner is operating or functioning properly and is not progressing toward any hardware malfunctions. In contrast, if any of those threshold values is unsatisfied, it can instead be concluded that, although the medical imaging scanner is currently operating or functioning acceptably, it is progressing toward a respective type of hardware malfunction (e.g., if the measured cooling fan speed does not satisfy the speed threshold, it can be concluded that the cooling fan is likely to fail in some threshold amount of time; if the measured heat exchanger outlet temperature does not satisfy the temperature threshold, it can be concluded that the heat exchanger is likely to fail in some threshold amount of time; if the measured cathode voltage does not satisfy the voltage threshold, it can be concluded that the cathode is likely to fail in some threshold amount of time). Unfortunately, existing techniques belonging to the threshold category can forecast only relatively simple types of hardware malfunctions, due to the high level of complex hardware interdependency involved in the medical imaging scanner. Indeed, the constituent pieces of hardware that make up the medical imaging scanner are often interconnected or interdependent on each other in ways that are unclear, obscure, or even unintended. Thus, although some malfunctions that the medical imaging scanner might experience can starkly present or manifest themselves mono-symptomatically and thus are easily diagnosable via threshold comparisons (e.g., an X-ray tube filament that is nearing a burn-out state can be easily detected when its measured electrical resistance sharply rises), many malfunctions that the medical imaging scanner might experience can instead subtly present or manifest themselves poly-symptomatically and thus are extremely difficult or impossible to diagnose via threshold comparisons (e.g., slip ring contamination might cause small perturbations in certain measured voltages or

currents but not in various measured temperatures or pressures; or slip ring contamination might instead cause small perturbations in those measured temperatures or pressures but not in the certain measured voltages or currents). It can be infeasible to exhaustively pinpoint beforehand which combinations of which sensor threshold values indicate which of those poly-symptomatic malfunctions, and so existing techniques belonging to the threshold category are useful only for forecasting very simple types of malfunctions, notably those where the physics of failure are well understood.

[0100] On the other hand, existing techniques belonging to the machine learning category forecast the future time at which excessive degradation is expected to occur by feeding all of the measured physical attributes of the constituent hardware of the medical imaging scanner as one collective input vector to a machine learning model (e.g., artificial neural network), and the machine learning model predicts as output an identity of whichever hardware malfunction that the machine learning model believes that the medical imaging scanner is likely to experience at the future time. In other words, the machine learning model can be considered as having been trained to recognize which patterns exhibited by the measured physical attributes of the medical imaging scanner are indicative of the imminency of which specific hardware malfunctions. Note that existing techniques belonging to the machine learning category can outperform existing techniques belonging to the threshold category. After all, the machine learning model is often trained in supervised fashion using sensor data vectors that are paired with ground-truth malfunction identities, meaning that there is no need for a technician or field engineer to exhaustively identify or define which combinations of which sensor threshold values are indicative of which hardware malfunctions. Instead, the machine learning model can be considered as implicitly determining those thresholds on its own during its training (e.g., training is an iterative procedure by which the modifiable internal weight matrices, convolutional kernels, or bias values of the machine learning model are incrementally optimized via backpropagation to increase a likelihood of producing “correct” outputs for any given inputs). Unfortunately, however, existing techniques belonging to the machine learning category often suffer from lack of generalizability due to scarce training data. Indeed, the machine learning model can reliably identify only hardware malfunctions on which it has been sufficiently trained (e.g., only hardware malfunctions many examples of which the machine learning model encountered during training). Training data is generally scarce in the medical or clinical context, due to the generally relatively small number of systems (e.g. a few hundred of each type) and the even smaller number of failure instances for each given failure mode (e.g. a dozen), and sometimes also due to patient privacy concerns. However, this scarcity is exacerbated for hardware malfunctions of medical imaging scanners, since such malfunctions are infrequent in the real-world. After all, equipment manufacturers, hospitals, and medical professionals go to great lengths to prevent medical imaging scanners from breaking down. So, for any given hardware malfunction, there might only be a small number (e.g., single digit) of examples of such given hardware malfunction occurring in the real-world. In other words, there might only be a few sensor data vectors that are annotated or otherwise labeled with the identity of that given hardware malfunction. Accordingly,

such given hardware malfunction can be considered as not being well-represented within whatever training data is used to train the machine learning model, thereby causing the machine learning model to be unable to reliably detect or predict the given hardware malfunction. Because many, if not most, of the hardware malfunctions which the medical imaging scanner might possibly suffer can be considered as rare or infrequent in this way, the predictive usefulness or generalizability of the machine learning model can be considered as being severely restricted.

[0101] Fortunately, various embodiments described herein can address the shortcomings that plague existing techniques for facilitating predictive maintenance of medical imaging scanners. As explained at length above, various embodiments described herein facilitate predictive maintenance of medical imaging scanners by leveraging vector clustering in a transformed feature space. More specifically, rather than predicting maintenance directly from the measured physical attributes of the constituent hardware of the medical imaging scanner (e.g., directly from **202** via machine learning or via thresholding like existing techniques), various embodiments described herein instead transform (e.g., by **210**) such measured physical attributes (or statistical metrics computed (e.g., by **208**) therefrom) into a compressed feature space (e.g., such as illustrated by **310**) and subsequently predict maintenance based on which clusters (e.g., **402**, **404**, **406**, **408**) of ground-truth data to which such transformed data is closest (e.g., in terms of Euclidean distance or cosine similarity). Such embodiments do not require exhaustive identification of sensor data thresholds, which means that such embodiments can be used to predict maintenance for the complicated, poly-symptomatic malfunctions that confound existing techniques belonging to the threshold category. Additionally, although such embodiments do rely on having at least some sensor data vectors that are paired, labeled, or annotated with ground-truth malfunction identities, such embodiments apply transformation and clustering to those annotated sensor data vectors rather than utilizing those annotated sensor data vectors for iterative training. Note that a cluster representing a given malfunction identity can be confidently defined by as few as one single sensor data vector that is annotated with that given malfunction identity. In stark contrast, a machine learning model cannot learn to confidently recognize that given malfunction identity by being trained only on one single sensor data vector that is annotated with that given malfunction identity. Thus, such embodiments do not suffer restricted generalizability caused by limited amounts of training data, unlike existing techniques belonging to the machine learning category. Accordingly, predicting maintenance based on sensor data transformation and clustering as described herein can be considered as outperforming existing techniques.

[0102] FIGS. 11-13 help to clarify or elaborate upon how various embodiments described herein can outperform existing techniques.

[0103] First, consider FIG. 11. In various aspects, the medical imaging device **120** can, as explained above, be any suitable type of scanning equipment for producing scanned medical images of patients (e.g., humans, animals, or otherwise). As some non-limiting examples, the medical imaging device **120** can be: a CT scanner; an MRI scanner; an X-ray scanner; an ultrasound scanner; a PET scanner; or an NM scanner. In any case, the medical imaging device **120** can be equipped or outfitted with a plurality of sensors **1102**.

In various instances, the plurality of sensors **1102** can include m sensors (e.g., one sensor per sensor parameter described above): a sensor **1102(1)** to a sensor **1102(m)**. In various aspects, each of the plurality of sensors **1102** can be any suitable electrical or electromechanical sensor that monitors any suitable measurable physical attribute exhibited by a respective piece of constituent hardware of the medical imaging device **120**. Some non-limiting examples of such measurable physical attributes can include: temperature exhibited by any piece of constituent hardware (e.g., any of the plurality of sensors **1102** can be a thermocouple or thermistor); pressure experienced by any piece of constituent hardware (e.g., any of the plurality of sensors **1102** can be a pressure transducer); deflection exhibited by any piece of constituent hardware (e.g., any of the plurality of sensors **1102** can be a strain gauge); a linear or angular displacement exhibited by any piece of constituent hardware (e.g., any of the plurality of sensors **1102** can be a linear or rotary potentiometer); linear or angular velocity exhibited by any piece of constituent hardware (e.g., any of the plurality of sensors **1102** can be velocimeter or tachometer); linear or angular acceleration exhibited by any piece of constituent hardware (e.g., any of the plurality of sensors **1102** can be an accelerometer); electric voltage exhibited by any piece of constituent hardware (e.g., any of the plurality of sensors **1102** can be a voltmeter); electric current exhibited by any piece of constituent hardware (e.g., any of the plurality of sensors **1102** can be an ammeter); electric resistance, capacitance, or inductance exhibited by any piece of constituent hardware (e.g., any of the plurality of sensors **1102** can be an LCR meter); or decibel level exhibited by any piece of constituent hardware (e.g., any of the plurality of sensors **1102** can be a microphone). Some non-limiting examples of constituent hardware of the medical imaging device **120** can include: an X-ray tube or any portion thereof (e.g., cathode, anode, insulator, or filament); an X-ray detector or any portion thereof (e.g., scintillator, photoconductor, photodiode array, electrode layer, or transistor array); a fluid pump or any portion thereof (e.g., impeller, rotor, stator, or bearings); a plenum or any portion thereof (e.g., fluid inlet, fluid outlet, heat exchanger fins or channels, fluid filter, or cooling fan or blower); or a gantry or any portion thereof (e.g., motor, slip ring). Note that two or more of the plurality of sensors **120** can monitor different measurable physical attributes of the same piece of constituent hardware as each other (e.g., the sensor **1102(1)** can monitor temperature of an X-ray tube filament, whereas the sensor **1102(m)** can monitor electrical resistivity of the X-ray tube filament).

[0104] In any case, the electronic data that is captured or recorded by the plurality of sensors **1102** at any present instant or moment in time (or possibly across any present interval or span of time) can be referred to as a multivariate sensor data vector **1104**. In various aspects, the multivariate sensor data vector **1104** can be made up of a plurality of numerical readouts **1106** that respectively correspond in one-to-one fashion to the plurality of sensors **1102** and thus to the measurable physical attributes of the medical imaging device **120**. Since the plurality of sensors **1102** can include m sensors, the plurality of numerical readouts **1106** can likewise include m readouts: a numerical readout **1106(1)** to a numerical readout **1106(m)**. In various instances, each of the plurality of numerical readouts **1106** can be whatever electronic data that is recorded or captured by a respective one of the plurality of sensors **1102**. As a non-limiting

example, the numerical readout **1106(1)** can be whatever scalars, vectors, matrices, or tensors are measured or outputted by the sensor **1102(1)** at the present instant or moment in time (or across the present interval or span of time, as appropriate). As another non-limiting example, the numerical readout **1106(m)** can be whatever scalars, vectors, matrices, or tensors are measured or outputted by the sensor **1102(m)** at the present instant or moment in time. Because the plurality of numerical readouts **1106** can have a one-to-one correspondence with the measurable physical attributes of the medical imaging device **120**, each of the plurality of numerical readouts **1106** can be considered as representing a real-world, interpretable quantity (e.g., representing a measured temperature, a measured pressure, or a measured voltage). Note that it can be the case that m is very large (e.g., in the hundreds or thousands). For this reason, the term “multivariate” can be considered as appropriate (e.g., signifying the immense collective size or dimensionality of the plurality of numerical readouts **1106**). In other words, the multivariate sensor data vector **1102** can be considered as a large (e.g., hundred-element or thousand-element) representation of the operational state of the medical imaging device **120** at the given instant or moment in time.

[0105] In various aspects, a set of statistical metrics **1108** can be computed, calculated, or otherwise derived (e.g., by **208**) from the multivariate sensor data vector **1104**. In various instances, the set of statistical metrics **1108** can include a total of p metrics for any suitable positive integer p : a statistical metric **1108(1)** to a statistical metric **1108(p)**. In various cases, each of the set of statistical metrics **1108** can be any suitable numerical quantity that can be obtained by applying any suitable statistics formula or algorithm (e.g., mean computation, median computation, mode computation, quartile computation, variance computation, standard deviation computation) or physical science formula or algorithm (e.g., heat transfer coefficient computation, drag coefficient computation, thermal conductivity computation, stress computation, strain computation, fatigue computation, Reynolds number computation, Nusselt number computation, electrical resistance computation) to any of the plurality of numerical readouts **1106**. Thus, the set of statistical metrics **1108** can be considered as supplying additional context that supplements or complements the multivariate sensor data vector **1104**.

[0106] In various aspects, any of the plurality of numerical readouts **1106** or any of the set of statistical metrics **1108** can be based on any suitable temporal aggregations applied to the plurality of sensors **1102**. Indeed, different ones of the plurality of sensors **1102** might be configured to produce numerical readouts at different or respective temporal frequencies (e.g., a voltage sensor might measure voltage every millisecond, while a pressure sensor might measure pressure every second). Accordingly, it is possible that there is no single timestep or timestamp at which each of the plurality of numerical readouts **1106** is freshly measured. In order to account for this temporal mismatch, any suitable type of temporal aggregation can be applied. As a non-limiting example, consider a given temperature sensor of the plurality of sensors **1102**. In some cases, whichever one of the plurality of numerical readouts **1106** that corresponds to that given temperature sensor can be whatever temperature measurement was most recently captured by that given temperature sensor. In some instances, any of the set of statistical metrics **1108** can be an average of whatever temperature

measurements were captured by that given temperature sensor over a preceding time span (e.g., over the past 24 hours). In some aspects, any of the set of statistical metrics **1108** can be a slope exhibited whatever temperature measurements were captured by that given temperature sensor over a preceding time span (e.g., over the past 24 hours). In some cases, any of the set of statistical metrics **1108** can be a standard deviation or variance of whatever temperature measurements were captured by that given temperature sensor over a preceding time span (e.g., over the past 24 hours). In some instances, any of the set of statistical metrics **1108** can be a maximum or minimum of whatever temperature measurements were captured by that given temperature sensor over a preceding time span (e.g., over the past 24 hours).

[0107] In various aspects, the multivariate sensor data vector **1104** or the set of statistical metrics **1108** can be converted (e.g., by **210**) into a transformed vector **1110**. In various instances, such conversion can be facilitated via any suitable dimensionality-reduction technique. Non-limiting examples of dimensionality-reduction techniques can include: principal component analysis (PCA); t-distributed stochastic neighbor embedding (t-SNE); or uniform manifold approximation and projection (UMAP). In any case, the transformed vector **1110** can be made up of a plurality of numerical elements **1112**. In various aspects, the plurality of numerical elements **1112** can include a total of n elements for any suitable positive integer $n > 1$: a numerical element **1112(1)** to a numerical element **1112(n)**. In various instances, each of the plurality of numerical elements **1112** can be any suitable scalar. Note that it can be the case that $n = m$, p . Indeed, it can be the case that $n = 3$ or $n = 2$, whereas m can be in the hundreds or thousands and p can be in the tens or dozens. Accordingly, the plurality of numerical elements **1112** can be considered as not having or exhibiting a one-to-one correspondence with the measurable physical attributes of the medical imaging device **120**. Indeed, none of the plurality of numerical elements **1112** can represent a discrete, real-world, interpretable quantity (e.g., none of the plurality of numerical elements **1112** is a measured temperature, measured pressure, or measured voltage). Instead, each of the plurality of numerical elements **1112** can be considered as a latent coordinate into which some portion of the multivariate sensor data vector **1104** or of the set of statistical metrics **1108** is embedded. Thus, the transformed vector **1110** can be considered as a dimensionally-compressed, latent representation of the multivariate sensor data vector **1104** or of the set of statistical metrics **1108** (e.g., can be considered as occupying a much smaller numerical space than the multivariate sensor data vector **1104** or the set of statistical metrics **1108**). In other words, the transformed vector **1110** can be considered as a small (e.g., two-element or three-element) representation of the operational state of the medical imaging device **120** at the present instant or moment in time.

[0108] Next, consider FIG. 12. In various embodiments, there can be a plurality of clusters **1202**. In various aspects, the plurality of clusters **1202** can include a total of t clusters for any suitable positive integer $t > 1$: a cluster **1202(1)** to a cluster **1202(t)**. In various instances, each of the plurality of clusters **1202** can contain any suitable number of ground-truth transformed vectors, each ground-truth transformed vector having the same format, size, or dimensionality as the transformed vector **1110**. As a non-limiting example, if the

transformed vector **1110** is a 3-element vector, then each ground-truth transformed vector can likewise be a 3-element vector. In any case, each ground-truth transformed vector can be computed (e.g., as described with respect to FIG. 11) from a respective multivariate sensor data vector which was measured by some instantiation of the plurality of sensors **1102** for some instantiation of the medical imaging device **120** that was known or deemed to be suffering from some respective hardware malfunction. In particular, all ground-truth transformed vectors belonging to the same cluster as each other can be considered as corresponding to or being labeled or annotated with the same hardware malfunction as each other.

[0109] As a non-limiting example, the cluster **1202(1)** can include a total of q_1 vectors for any suitable positive integer q_1 : a ground-truth transformed vector **1202(1)(1)** to a ground-truth transformed vector **1202(1)(q₁)**. In various aspects, each ground-truth transformed vector in the cluster **1202(1)** can correspond to or be labeled with a first hardware malfunction (e.g., X-ray tube arcing). For instance, there can have been a first instantiation or copy of the medical imaging device **120** (e.g., located in some first hospital or clinic) that was known or deemed to be suffering from that first hardware malfunction at some particular point in time (or over some particular interval of time); the sensors with which that first instantiation or copy of the medical imaging device **120** was equipped can have readout a first multivariate sensor data vector (e.g., having the same format, size, or dimensionality as the multivariate sensor data vector **1104**) at that particular point in time; and that first multivariate sensor data vector can have been converted into the ground-truth transformed vector **1202(1)(1)** as described with respect to FIG. 11. Thus, the ground-truth transformed vector **1202(1)(1)** can be considered as a low-dimensional latent representation of the operational state of that first instantiation or copy of the medical imaging device **120** when that first instantiation or copy was known to be afflicted with the first hardware malfunction. Likewise, there can have been a q_1 -th instantiation or copy of the medical imaging device **120** (e.g., located in some q_1 -th hospital or clinic) that was known or deemed to be suffering from that first hardware malfunction at some certain point in time (or over some certain interval of time); the sensors with which that q_1 -th instantiation or copy of the medical imaging device **120** was equipped can have readout a q_1 -th multivariate sensor data vector at that certain point in time; and that q_1 -th multivariate sensor data vector can have been converted into the ground-truth transformed vector **1202(1)(q₁)** as described with respect to FIG. 11. So, the ground-truth transformed vector **1202(1)(q₁)** can be considered as a low-dimensional latent representation of the operational state of that q_1 -th instantiation or copy of the medical imaging device **120** when that q_1 -th instantiation or copy was known to be afflicted with the first hardware malfunction.

[0110] As another non-limiting example, the cluster **1202(t)** can include a total of q_t vectors for any suitable positive integer q_t : a ground-truth transformed vector **1202(t)(1)** to a ground-truth transformed vector **1202(t)(q_t)**. In various aspects, each ground-truth transformed vector in the cluster **1202(t)** can correspond to or be labeled with a t -th hardware malfunction (e.g., gantry bearing failure). For instance, there can have been a first instantiation or copy of the medical imaging device **120** that was known or deemed to be suffering from that t -th hardware malfunction at some par-

ticular point in time; the sensors with which that first instantiation or copy of the medical imaging device **120** was equipped can have readout a first multivariate sensor data vector at that particular point in time; and that first multivariate sensor data vector can have been converted into the ground-truth transformed vector $\mathbf{1202}(t)(1)$ as described with respect to FIG. 11. Thus, the ground-truth transformed vector $\mathbf{1202}(t)(1)$ can be considered as a low-dimensional latent representation of the operational state of that first instantiation or copy of the medical imaging device **120** when that first instantiation or copy was known to be afflicted with the t -th hardware malfunction. Likewise, there can have been a q_t -th instantiation or copy of the medical imaging device **120** that was known or deemed to be suffering from that t -th hardware malfunction at some certain point in time; the sensors with which that q_t -th instantiation or copy of the medical imaging device **120** was equipped can have readout a q_t -th multivariate sensor data vector at that certain point in time; and that q_t -th multivariate sensor data vector can have been converted into the ground-truth transformed vector $\mathbf{1202}(t)(q_t)$ as described with respect to FIG. 11. So, the ground-truth transformed vector $\mathbf{1202}(t)(q_t)$ can be considered as a low-dimensional latent representation of the operational state of that q_t -th instantiation or copy of the medical imaging device **120** when that q_t -th instantiation or copy was known to be afflicted with the t -th hardware malfunction.

[0111] Note that, in some cases, one of the plurality of clusters **1202** can correspond to no hardware malfunction at all (e.g., can correspond to normal, expected, or healthy operation of the medical imaging device **120**), and the rest or remainder of the plurality of clusters **1202** can correspond to respective or distinct hardware malfunctions.

[0112] Furthermore, note that, although ground-truth transformed vectors can be organized into clusters (e.g., into **1202**) based on their known hardware malfunction labels (e.g., first the labels are known, and all vectors having the same label can subsequently be deemed to be in the same cluster as each other), this is a mere non-limiting example for ease of explanation. In other embodiments, the hardware malfunction labels might not yet be known. In such cases, the ground-truth transformed vectors can be organized into clusters using any suitable clustering algorithm (e.g., **306**), such as K-means clustering or fuzzy clustering, and malfunction labels can be respectively applied to the clusters later.

[0113] In any case, note that, because the real-world frequency or occurrence rate of medical imaging hardware malfunctions can be low as explained above, it can be possible that q_t is small in magnitude (e.g., q_t is in the tens or even signal digits) for any or all integers $1 \leq t \leq T$. In other words, it can be the case that there are not many or voluminous real-world examples of some hardware malfunctions afflicting the medical imaging device **120** (or instantiations thereof). Accordingly, if a machine learning model were trained on the plurality of clusters **1202** to predict hardware malfunction identities in response to inputted transformed vectors (or in response to inputted multivariate sensor data vectors), such machine learning model would not achieve high prediction accuracy or precision. That is, if such a machine learning model were to be executed post-training on the transformed vector **1110** (or on the multivariate sensor data vector **1104**), such machine learning model would not be able to confidently determine

which (if any) hardware malfunction the medical imaging device **120** is likely to soon suffer.

[0114] To overcome this problem of limited training data, the clustered organization of the ground-truth transformed vectors can be leveraged. In particular, a plurality of deviations **1204** can be computed (e.g., by **112**) between the transformed vector **1110** and the plurality of clusters **1202**. As a non-limiting example, a deviation $\mathbf{1204}(1)$ can be equal to or otherwise based on any suitable error metric (e.g., Euclidean distance, cross-entropy error, mean absolute error, mean squared error, reciprocal of cosine similarity) between the transformed vector **1110** and the centroid of the cluster **1202(1)**. Thus, the deviation $\mathbf{1204}(1)$ can be considered as indicating as how far or how different the transformed vector **1110** is to the average transformed vector in the cluster **1202(1)**. As another non-limiting example, a deviation $\mathbf{1204}(t)$ can be equal to or otherwise based on any suitable error metric between the transformed vector **1110** and the centroid of the cluster $\mathbf{1202}(t)$. So, the deviation $\mathbf{1204}(t)$ can be considered as indicating as how far or how different the transformed vector **1110** is to the average transformed vector in the cluster $\mathbf{1202}(t)$.

[0115] In various cases, the deviation $\mathbf{1204}(1)$ to the deviation $\mathbf{1204}(t)$ can collectively be considered as the plurality of deviations **1204**.

[0116] In various aspects, the plurality of deviations **1204** can be considered as indicating which (if any) hardware malfunction the medical imaging device **120** is likely to suffer, and a maintenance or servicing task can be recommended (e.g., by **114**) as appropriate. As a non-limiting example, the transformed vector **1110** can be considered as belonging to whichever one of the plurality of clusters **1202** from whose centroid the transformed vector **1110** is least deviated (e.g., if the transformed vector **1110** is closest to the centroid of the cluster corresponding to normal operation, then it can be concluded that the medical imaging device **120** is not likely to suffer any hardware malfunctions soon, and so no maintenance or servicing task can be recommended; if the transformed vector **1110** is closest to the centroid of the cluster corresponding to X-ray tube arcing, then it can be concluded that the medical imaging device **120** is likely to suffer X-ray tube arcing soon, and so an X-ray tube replacement can be recommended). As another non-limiting example, the transformed vector **1110** can be considered as belonging to whichever ones of the plurality of clusters **1202** from whose centroids the transformed vector **1110** is deviated by at least any suitable threshold margin or amount (e.g., if the transformed vector **1110** is within a threshold deviation of the centroid of the cluster corresponding to slip ring contamination, then it can be concluded that the medical imaging device **120** is likely to suffer slip ring contamination soon, and so slip ring cleaning can be recommended; if the transformed vector **1110** is also within a threshold deviation of the centroid of the cluster corresponding to X-ray tube arcing, then it can be concluded that the medical imaging device **120** is likely to also suffer slip ring contamination soon, and so slip ring cleaning can also be recommended).

[0117] In this way, whatever specific hardware malfunctions that the medical imaging device **120** is likely to suffer in the near future can be reliably predicted, notwithstanding the fact that the plurality of clusters **1202** might not be voluminous enough to sufficiently train a machine learning

model (e.g., sufficient training often requires thousands, hundreds of thousands, or even millions of input-annotation pairs).

[0118] Note that, in various aspects, each of the plurality of deviations **1204** can be computed in timeseries fashion (e.g., can be computed each month, week, day, hour, minute, or second) and therefore can be considered as a respective one of the above-mentioned degradation scores or degradation indices.

[0119] Additionally, note that, in response to predicting that the medical imaging device **120** is likely to suffer a particular hardware malfunction in the near future, various embodiments described herein can automatically recommend (e.g., via **114**) that an appropriate maintenance or servicing task be performed within any suitable threshold amount of time on the medical imaging device **120**, but various embodiments described herein can also perform (e.g., via **114**) any other suitable automated actions in response to such prediction. As a non-limiting example, various embodiments described herein can automatically schedule (e.g., via **114**) that appropriate maintenance or servicing task (e.g., automatically populate that task on an electronic calendar of a field engineer or technician). As another non-limiting example, various embodiments described herein can automatically disable (e.g., via **114**) the medical imaging device **120** (e.g., automatically prohibit or prevent the medical imaging device **120** from performing new scans) until the appropriate maintenance or servicing task has been performed. This can be particularly useful, since operating with degraded hardware can exacerbate or worsen damage to the medical imaging device **120**. As still another non-limiting example, various embodiments described herein can automatically reduce an operation of the medical imaging device **120** to a less rigorous state until the appropriate maintenance or servicing task has been performed (e.g., automatically permit the medical imaging device **120** to perform new scans at no more than a limited level of power or limited dosage of radiation). This can be particularly useful, since fully operating with degraded hardware can potentially expose medical patients to dangerous levels of electricity or radiation. As yet another non-limiting example, various embodiments described herein can automatically cause the medical imaging device **120** to run or initiate any suitable self-diagnostic, self-troubleshooting, or self-healing protocols (e.g., automatically estimating how much time remains until the particular hardware malfunction occurs by running a self-diagnostic test and automatically recommending or scheduling the appropriate maintenance or servicing task to be performed prior to that estimated time; automatically rebooting the medical imaging device **120**; automatically resetting the medical imaging device **120** to default or factory configuration settings). In some cases, if those self-diagnostic, self-troubleshooting, or self-healing protocols solve the particular hardware malfunction (e.g., if a post-reboot or post-reset version of the transformed vector **1110** is now closest to the normal or standard operation cluster), then the recommendation or scheduling of the appropriate maintenance or servicing task can be cancelled (e.g., such task can be no longer needed). Instead, if those self-diagnostic, self-troubleshooting, or self-healing protocols fail to solve the particular hardware malfunction (e.g., if a post-reboot or post-reset version of the transformed vector **1110** is still not closest to the normal or standard operation cluster), then the recommendation or

scheduling of the appropriate maintenance or servicing task can be resumed or emphasized (e.g., such task can be still needed). As even another non-limiting example, various embodiments described herein can visually render on any suitable electronic screen or display associated with the medical imaging device **120** a tutorial (e.g., step-by-step instructions) that explains how to perform the appropriate maintenance or servicing task.

[0120] Note that, although it is feasible to perform clustering directly on the multivariate sensor data vectors from which the ground-truth transformed vectors (e.g., making up **1202**) are derived, such an embodiment would not be optimal. Indeed, because each multivariate sensor data vector can be of such immense dimensionality, applying clustering and deviation computations directly to such multivariate sensor data vectors would be severely computationally intensive (e.g., would consume excessive amounts of computer memory, computer processing power, or computation time). These excessive computational costs are greatly reduced by transforming such multivariate sensor data vectors into a low-dimensional latent space prior to performing such clustering and deviation computations. In other words, the multivariate sensor data vectors described herein (e.g., the raw sensor data) can be considered as extremely unwieldy, bulky, or cumbersome, whereas the transformed vectors described herein can instead be considered as lithe, light-weight, or manageable.

[0121] As mentioned above, any of the plurality of numerical readouts **1106** or any of the set of statistical metrics **1108** can be based on any suitable types of temporal aggregations applied to the plurality of sensors **1102**. In some aspects, the herein-described transformation and clustering can be considered as pruning away unimportant or unhelpful ones of those temporal aggregations (e.g., as discarding temporal aggregations that do not carry predictive weight or add a meaningful dimension with respect to hardware malfunctions of the medical imaging device **120**). Thus, a user, technician, or medical professional can freely define or incorporate whatever temporal aggregations they desire into the plurality of numerical readouts **1106** or into the set of statistical metrics **1108** without having to worry about whether or not such aggregations will be helpful or dispositive for maintenance prediction, since the subsequent transformation and clustering can be considered as eliminating or not paying attention to any aggregations that turn out to be unhelpful (e.g., that turn out to not carry predictive weight or add a meaningful dimension with respect to hardware malfunctions).

[0122] Now, consider FIG. 13, which shows a computer-implemented method **1300** for facilitating various embodiments described herein.

[0123] In various aspects, act **1302** can involve reading a multivariate sensor data vector (e.g., **1104**) measured by a plurality of electronic sensors (e.g., **1102**) that monitor constituent hardware of a medical imaging scanner (e.g., **120**), wherein there can be a one-to-one correspondence between numerical elements (e.g., **1106**) of the multivariate sensor data vector and measurable physical attributes exhibited by the constituent hardware of the medical imaging scanner.

[0124] In various instances, act **1304** can involve converting, via execution of a dimensionality-reduction technique, the multivariate sensor data vector or statistical metrics (e.g., **1108**) derived therefrom into a transformed vector (e.g.,

1110) whose dimensionality is smaller than that of the multivariate sensor data vector or of the statistical metrics derived therefrom, wherein there can be no one-to-one correspondence between numerical elements (e.g., 1112) of the transformed vector and the measurable physical attributes exhibited by the constituent hardware of the medical imaging scanner.

[0125] In various cases, act 1306 can involve respectively computing a set of deviations (e.g., 1204) between: the transformed vector; and centroids of a set of ground-truth transformed vector clusters (e.g., 1202), wherein one of the set of ground-truth transformed vector clusters represents normal operation of the medical imaging scanner, and wherein each remaining one of the set of ground-truth transformed vector clusters represents a respective hardware malfunction of the medical imaging scanner.

[0126] In various aspects, act 1308 can involve identifying which, if any, hardware malfunction the medical imaging scanner is likely to suffer, based on the set of deviations.

[0127] In various instances, act 1310 can include adjusting operation of the medical imaging scanner based on the identified hardware malfunction (e.g., rebooting, disabling, permitting only limiting operation).

[0128] Various embodiments described herein can integrate into a practical application various teachings relating to the field of medical imaging. As described above, schedule-based maintenance of medical imaging scanners can often lead to over-servicing (which is wasteful) or under-servicing (which is bad for patient outcomes). Existing techniques attempt to rectify this by predicting which piece of hardware will suffer from excessive degradation at some point in the near future and proactively servicing such piece of hardware shortly before that point in the near future. Some existing techniques do this by comparing each sensor readout to a respective threshold. Unfortunately, those existing techniques can only predict relatively simple types of hardware malfunctions due to the impracticality of exhaustively identifying interdependent threshold values for complicated, poly-symptomatic malfunctions. Other existing techniques do this by feeding all of the sensor readouts as a single input vector to a machine learning model. Unfortunately, those other existing techniques are severely limited by lack of voluminous training data. Thus, existing techniques can be considered as suffering from various technical problems.

[0129] Various embodiments described herein can address one or more of these technical problems. In particular, the present inventors realized that the disadvantages of existing techniques can be overcome by compressing raw sensor data into a transformed feature space and by applying clustering to that transformed feature space. More specifically, the raw data readouts produced by the hardware sensors of a medical imaging scanner can be collectively considered as forming a multivariate vector of high dimensionality (e.g., having hundreds of numerical elements). That multivariate vector (and any statistical metrics derived from it) can be compressed via dimensionality-reduction (e.g., PCA, t-SNE, UMAP) into a transformed vector that is much smaller (e.g., having only two or three numerical elements). Such dimensionality-reduction can be considered as converting the multivariate vector (or its statistical metrics) from: an unwieldy or cumbersome format to a handy, light-weight format. Now, a respective ground-truth transformed vector can be obtained in this way for any medical imaging

scanners that are known or deemed to be suffering from specific hardware malfunctions, hence the term “ground-truth” (e.g., those transformed vectors can be considered as being labeled or annotated with the identities of those specific hardware malfunctions). Accordingly, the transformed feature space can be populated with clusters of ground-truth transformed vectors, each cluster representing scanners that were known or deemed to be suffering from some respective hardware malfunction. By computing respective deviations (e.g., Euclidean distances) between the centroids of those clusters and the transformed vector of the medical imaging scanner in question, it can be determined which hardware malfunction the medical imaging scanner already suffers or will shortly suffer from. Note that such embodiments do not require exhaustive identification of interdependent sensor thresholds, unlike existing techniques that are based on thresholding. Thus, such embodiments can be applied even to highly complex or complicated types of hardware malfunctions. Additionally, note that such embodiments do not involve using the ground-truth transformed vectors for machine learning model training, unlike existing techniques that are based on machine learning. Instead, such embodiments utilize the ground-truth transformed vectors for clustering and deviation computation, which can be reliably performed even for clusters that are small in size (e.g., even for clusters that have only one or two vectors in them). Accordingly, such embodiments do not suffer from restricted generalizability caused by lack of training data.

[0130] In other words, the herein-described clustering and centroidal deviations can be considered as avoiding the problems of exhaustive threshold definition and training data scarcity that plague existing techniques, and the herein-described transformation into a low-dimensional latent space can be considered as ensuring that the computational cost of such clustering and centroidal deviation computations remains manageable or practical.

[0131] It must be emphasized how counter-intuitive various embodiments described herein are. The herein-described embodiments involve two primary features: (1) transformation of high-dimensional sensor data into a low-dimensional latent space; and (2) computing deviations with respect to cluster centroids within that low-dimensional latent space. As mentioned above, computing centroid cluster deviations without first transforming into the low-dimensional latent space would be prohibitively computationally expensive. Thus, one of ordinary skill in the art would not have even entertained the idea of implementing (2) without (1). However, one of ordinary skill in the art would not even have considered implementing (1) at all, since it would have been expected to be incompatible or counter-productive with respect to existing techniques. Indeed, for existing techniques belonging to the threshold category, transformation of sensor data into the low-dimensional latent space would have made the already-difficult job of identifying appropriate threshold values exponentially harder. This is because the numerical coordinates in the low-dimensional latent space have no real-world, physical, or interpretable meanings. In other words, because it was already hard enough for technicians or field engineers to identify appropriate thresholds for measurable physical attributes about which they have real-world intuition (e.g., like temperature, pressure, or voltage), it would have been significantly more difficult if not impossible for technicians or field engineers to identify appropriate thresholds for latent coordinates that have no

readily interpretable or understandable correspondence with such measurable physical attributes (e.g., latent coordinates can appear or seem to be random or meaningless numbers; they are not temperatures, pressures, or voltages; technician or field engineers have no physical intuitive about the behaviors of such latent coordinates). Additionally, for existing techniques belonging to the machine learning category, transformation of sensor data into the low-dimensional latent space would have been viewed as tantamount to discarding potentially valuable input information. After all, the animating principle that underlies the state-of-the-art of machine learning is that more input data is better (e.g., more input data being fed to a machine learning model provides more context, which allows for the machine learning model to make more informed decisions). Although dimensionality-reduction techniques aim to preserve substantive content of whatever data is being compressed, such dimensionality-reduction techniques invariably involve information loss. Indeed, applying dimensionality-reduction to a vector can be thought somewhat as retaining elements or degrees of freedom of the vector that seem important while throwing away elements or degrees of freedom of the vector that seem unimportant (e.g., such importance or unimportance can be defined or measured differently depending on which specific dimensionality-reduction technique is used). In any case, because of this information loss, one of ordinary skill in the art would not have contemplated feeding transformed, low-dimensional latent vectors instead of the multivariate sensor data as input to a machine learning model configured to predict medical imaging hardware malfunctions. After all, the person of ordinary skill in the art would have considered doing so to be tantamount to throwing away potentially valuable context and therefore would have been expected to reduce prediction accuracy or reliability of the machine learning model (e.g., giving the machine learning model less data would be expected to make the machine learning model's decision less informed). Therefore, various embodiments described herein can be considered as a clever, unusual, surprising, or counter-intuitive way of predicting hardware degradation for medical imaging scanners, when considered against the backdrop of existing techniques.

[0132] For at least these reasons, various embodiments described herein certainly constitute a tangible and concrete technical improvement, technical effect, or technical advantage in the field of medical imaging. Accordingly, such embodiments clearly qualify as useful and practical applications of computers.

What is claimed is:

1. A system comprising:

a processor that executes computer-executable components stored in a non-transitory computer-readable memory, wherein the computer-executable components comprise:

an input component that reads a multivariate sensor data vector measured by a plurality of electronic sensors that monitor constituent hardware of a medical imaging scanner, wherein there is a one-to-one correspondence between numerical elements of the multivariate sensor data vector and measurable physical attributes exhibited by the constituent hardware of the medical imaging scanner;

a degradation component that:

converts, via execution of a dimensionality-reduction technique, the multivariate sensor data vector

or statistical metrics derived therefrom into a transformed vector whose dimensionality is smaller than that of the multivariate sensor data vector or of the statistical metrics derived therefrom, wherein there is no one-to-one correspondence between numerical elements of the transformed vector and the measurable physical attributes exhibited by the constituent hardware of the medical imaging scanner; and

respectively computes a set of deviations between: the transformed vector; and centroids of a set of ground-truth transformed vector clusters, wherein one of the set of ground-truth transformed vector clusters represents normal operation of the medical imaging scanner, and wherein each remaining one of the set of ground-truth transformed vector clusters represents a respective hardware malfunction of the medical imaging scanner; and

a planned maintenance component that:

determines that the transformed vector belongs to whichever one of the set of ground-truth transformed vector clusters from whose centroid the transformed vector is least deviated; and

in response to determining that the transformed vector belongs to a first ground-truth transformed vector cluster that represents a first hardware malfunction, recommends that a maintenance task corresponding to the first hardware malfunction be performed on the medical imaging scanner.

2. The system of claim 1, wherein the planned maintenance component disables the medical imaging scanner until the maintenance task is performed.

3. The system of claim 1, wherein the planned maintenance component permits the medical imaging scanner to operate only at a limited radiation dosage until the maintenance task is performed.

4. The system of claim 1, wherein the planned maintenance component:

estimates, via a self-diagnostic test, a time remaining until the first hardware malfunction occurs; and

recommends that the maintenance task be performed prior to the estimated time.

5. The system of claim 1, wherein the planned maintenance component:

reboots the medical imaging scanner or a portion thereof; and

recommends that the maintenance task be performed, in response to a post-reboot transformed vector of the medical imaging scanner belonging to the first ground-truth transformed vector cluster.

6. The system of claim 1, wherein the planned maintenance component renders on an electronic screen a tutorial explaining how to perform the maintenance task on the medical imaging scanner.

7. The system of claim 1, wherein the planned maintenance component automatically schedules the maintenance task for performance on the medical imaging scanner.

8. The system of claim 1, wherein the measurable physical attributes of a respective piece of constituent hardware of the medical imaging scanner include a temperature of the respective piece of constituent hardware, a pressure of the respective piece of constituent hardware, a deflection of the respective piece of constituent hardware, a linear or angular displacement of the respective piece of constituent hard-

ware, a linear or angular velocity of the respective piece of constituent hardware, a linear or angular acceleration of the respective piece of constituent hardware, an electrical voltage or electrical current of the respective piece of constituent hardware, or a decibel level of the respective piece of constituent hardware.

9. The system of claim 1, wherein the constituent hardware of the medical imaging scanner includes an X-ray tube, an X-ray detector, a fluid pump, a filter, a heat exchanger, a fan, a plenum, or a gantry.

10. A computer-implemented method comprising:

reading, by a device operatively coupled to a processor, a multivariate sensor data vector measured by a plurality of electronic sensors that monitor constituent hardware of a medical imaging scanner, wherein there is a one-to-one correspondence between numerical elements of the multivariate sensor data vector and measurable physical attributes exhibited by the constituent hardware of the medical imaging scanner;

converting, by the device and via execution of a dimensionality-reduction technique, the multivariate sensor data vector or statistical metrics derived therefrom into a transformed vector whose dimensionality is smaller than that of the multivariate sensor data vector or of the statistical metrics derived therefrom, wherein there is no one-to-one correspondence between numerical elements of the transformed vector and the measurable physical attributes exhibited by the constituent hardware of the medical imaging scanner;

respectively computing, by the device, a set of deviations between: the transformed vector; and centroids of a set of ground-truth transformed vector clusters, wherein one of the set of ground-truth transformed vector clusters represents normal operation of the medical imaging scanner, and wherein each remaining one of the set of ground-truth transformed vector clusters represents a respective hardware malfunction of the medical imaging scanner;

determining, by the device, that the transformed vector belongs to whichever one of the set of ground-truth transformed vector clusters from whose centroid the transformed vector is least deviated; and

recommending, by the device and in response to determining that the transformed vector belongs to a first ground-truth transformed vector cluster that represents a first hardware malfunction, that a maintenance task corresponding to the first hardware malfunction be performed on the medical imaging scanner.

11. The computer-implemented method of claim 10, further comprising:

disabling, by the device, the medical imaging scanner until the maintenance task is performed.

12. The computer-implemented method of claim 10, further comprising:

permitting, by the device, the medical imaging scanner to operate only at a limited radiation dosage until the maintenance task is performed.

13. The computer-implemented method of claim 10, further comprising:

estimating, by the device and via a self-diagnostic test, a time remaining until the first hardware malfunction occurs; and

recommending, by the device, that the maintenance task be performed prior to the estimated time.

14. The computer-implemented method of claim 10, further comprising:

rebooting, by the device, the medical imaging scanner or a portion thereof; and

recommending, by the device, that the maintenance task be performed, in response to a post-reboot transformed vector of the medical imaging scanner belonging to the first ground-truth transformed vector cluster.

15. The computer-implemented method of claim 10, further comprising:

rendering, by the device and on an electronic screen, a tutorial explaining how to perform the maintenance task on the medical imaging scanner.

16. The computer-implemented method of claim 10, further comprising:

automatically scheduling, by the device, the maintenance task for performance on the medical imaging scanner.

17. The computer-implemented method of claim 10, wherein the measurable physical attributes of a respective piece of constituent hardware of the medical imaging scanner include a temperature of the respective piece of constituent hardware, a pressure of the respective piece of constituent hardware, a deflection of the respective piece of constituent hardware, a linear or angular displacement of the respective piece of constituent hardware, a linear or angular velocity of the respective piece of constituent hardware, a linear or angular acceleration of the respective piece of constituent hardware, an electrical voltage or electrical current of the respective piece of constituent hardware, or a decibel level of the respective piece of constituent hardware.

18. The computer-implemented method of claim 10, wherein the constituent hardware of the medical imaging scanner includes an X-ray tube, an X-ray detector, a fluid pump, a filter, a heat exchanger, a fan, a plenum, or a gantry.

19. A computer program product for enabling remote planned maintenance of medical imaging devices, the computer program product comprising a computer readable storage medium having program instructions embodied therewith, the program instructions executable by a processor to cause the processor to:

read a multivariate sensor data vector measured by a plurality of electronic sensors that monitor constituent hardware of a medical imaging scanner, wherein there is a one-to-one correspondence between numerical elements of the multivariate sensor data vector and measurable physical attributes exhibited by the constituent hardware of the medical imaging scanner;

convert, via execution of a dimensionality-reduction technique, the multivariate sensor data vector or statistical metrics derived therefrom into a transformed vector whose dimensionality is smaller than that of the multivariate sensor data vector or of the statistical metrics derived therefrom, wherein there is no one-to-one correspondence between numerical elements of the transformed vector and the measurable physical attributes exhibited by the constituent hardware of the medical imaging scanner;

respectively compute a set of deviations between: the transformed vector; and centroids of a set of ground-truth transformed vector clusters, wherein one of the set of ground-truth transformed vector clusters represents normal operation of the medical imaging scanner, and wherein each remaining one of the set of ground-truth

transformed vector clusters represents a respective hardware malfunction of the medical imaging scanner; determine that the transformed vector belongs to whichever one of the set of ground-truth transformed vector clusters from whose centroid the transformed vector is least deviated; and

recommend, in response to determining that the transformed vector belongs to a first ground-truth transformed vector cluster that represents a first hardware malfunction, that a maintenance task corresponding to the first hardware malfunction be performed on the medical imaging scanner.

20. The computer program product of claim **19**, wherein the program instructions are further executable to cause the processor to:

disable, or reduce a radiation dosage of, the medical imaging scanner until the maintenance task is performed.

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